SST 404 — ECONOMETRICS II

Comprehensive Lesson Notes: Simultaneous Equations, Identification and Estimation

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*This edition reproduces the whole of the downloaded module — the front matter and **all seven lessons**, which now occupy all 72 pages of the file (see Appendix A). The download carried two defects that had to be repaired: the lesson prose, the two-column iLearn activity plan and the mathematics were laid on top of one another, and **every formula had been broken into loose symbols** — subscripts placed on their own baselines, so that a line such as $C_t = \beta_1 + \beta_2 Y_t$ arrived as "YC … ttt … 21". The mathematics has therefore been **re-set in full**, in the standard notation of the subject, and each worked example re-derived so that it reproduces the results the module states. Appendix B lists what was rebuilt and where the module's own figures were kept as printed.*

Course Description

SST 404: Econometrics II — 3 credit hours. **Prerequisite:** SST 300.

Purpose of the course. The main aim of the course unit is to lay a solid theoretical background and enhance the ability to implement econometric techniques to critique empirical studies in economics.

Expected learning outcomes. At the end of the course students should be able to:

1. specify structural and reduced forms of econometric models and answer questions relating to economic issues of a country;
2. make adequate interpretations of estimation results of macroeconomic models which exhibit relationships in the economy;
3. work with real data to analyse, summarize and present results from an empirical analysis.

Course content. Nature of simultaneous equations; identification: order and rank conditions; instrumental variables; estimation of simultaneous equation models: structural and reduced form equations, two-stage least squares and GMM; types of estimators and their asymptotic properties; hypothesis testing.

Mode of delivery. Class lectures, discussions, problem solving, exercises.

Assessment. Continuous assessment 30%; final examination 70%.

Lesson Map

Lesson	Title	Sections in this edition
1	System of simultaneous equations	§1.1–1.7
2	Simultaneous equation bias	§2.1–2.6
3	Structural and reduced form equations	§3.1–3.7
4	The problem of identification	§4.1–4.6
5	Simultaneous equation estimation: indirect least squares	§5.1–5.8
6	The two-stage least squares method (2SLS)	§6.1–6.5
7	The instrumental variable method (IV)	§7.1–7.5

> *[The module's own contents pages list Lessons 6 and 7 the other way round — "6. The Instrumental Variable Method (IV)" and "7. The Two-Stage Least Squares Method (2SLS)" — but the body of the module teaches 2SLS as Lesson 6 and IV as Lesson 7, and the learning outcomes, examples and references follow that order. This edition follows the body. The same pages give the section numbers of Lesson 6 as starting at 6.3 and the e-tivities of Lesson 7 as "6.4"/"6.5"/"6.6"; those too have been corrected here, without moving any content.]*

LESSON ONE: SYSTEM OF SIMULTANEOUS EQUATIONS

1.1 Introduction

In SST 300 (Econometric Methods) we discussed single-equation regression models in univariate and multivariate forms. One of the assumptions of least-squares theory is that there is one-to-one causation between the endogenous variable Y and the exogenous variables X. If this assumption is not true — that is, if there is *double causation* in the relationships, so that the exogenous variable X also depends on the endogenous variable Y — then the assumption we had made in least-squares theory,

$$E(XU) = 0,$$

is violated and is no longer equal to zero.

In such a scenario, where there is double causation in a function, we cannot treat that function in isolation as a single-equation model: it belongs to a wider system of equations which describes the relationship among the relevant variables, $Y = f(X)$ and $X = f(Y)$. This wider system is a multi-equation model in which Y and X both appear as endogenous variables. A system describing the joint dependence of variables is called a **system of simultaneous equations**.

We study simultaneous equations on their own, despite the exhaustive discussion of single-equation models, because parameter estimation here has special features which are not available in single-equation models — for example, some explanatory variables are related to stochastic variables. Under such circumstances the ordinary least squares method results in **biased and inconsistent** estimators. We therefore have to devise methods that give unbiased and consistent estimators. The major topics of concern will be the dependence of the error term on the regressor(s), simultaneous-equation bias, and the structural and reduced forms of simultaneous-equation models.

1.2 Learning outcomes

By the end of this lesson the learner will be able to:

1. describe the structure of a system of simultaneous equations;
2. write the structural form of simultaneous equations into reduced forms.

1.3 Simultaneous equation relationships

Suppose we want to estimate the demand for food. We know that this demand depends on its price and on the consumer's income, so we write the demand function

$$Q_t = \beta_1 + \beta_2 P_t + \beta_3 Y_t + u_t, \qquad (1.1)$$

where Q_t is the quantity demanded, P_t the price, Y_t the consumer's income and u_t the error term, with the usual assumptions

$$E(u_t) = 0, \qquad E(u_t^2) = \sigma^2, \qquad E(u_t u_{t'}) = 0 \ \ (t \neq t'), \qquad E(u_t Y_t) = 0.$$

If we apply ordinary least squares to this equation we will get biased estimates, because even if the price P is treated as an independent variable in this model, there exists another equation in which the price P is the dependent variable; hence the price may be correlated with the error term, that is $E(P_t u_t) \neq 0$.

The price equation in the system is the supply (or price-determination) relation

$$P_t = \gamma_1 + \gamma_2 Q_t + v_t, \qquad (1.2)$$

with

$$E(v_t) = 0, \qquad E(v_t^2) = \sigma_v^2, \qquad E(v_t v_{t'}) = 0 \ \ (t \neq t'), \qquad E(v_t u_t) = 0.$$

Thus the quantity depends on the price and the price on the quantity. We call this scenario **double causation**. We solve these two equations simultaneously.

Example 1: Income determination model. Suppose we have an economic income determination model of the form

$$C_t = \beta_1 + \beta_2 Y_t + u_t \qquad (1.3)$$
$$Y_t = C_t + I_t \qquad (1.4)$$

where C_t and Y_t are endogenous variables; I_t is an exogenous variable; u_t represents the error term; β_1 and β_2 are the parameters of the system to be estimated; and t represents time. The two equations above represent the consumption function and the national income identity respectively.

The error term satisfies

$$E(u_t) = 0, \qquad E(u_t^2) = \sigma^2, \qquad E(u_t u_{t'}) = 0, \qquad E(u_t I_t) = 0.$$

If we were to apply ordinary least squares to estimate the parameters, we need to check whether the appropriate assumptions are satisfied. Clearly the assumptions of non-autocorrelation and homoscedasticity are satisfied from the above. The critical assumption is that of independence between the stochastic term and the regressor.

Consequences of simultaneous equations. Substitute equation (1.3) into (1.4):

$$Y_t = \beta_1 + \beta_2 Y_t + u_t + I_t,$$

so that

$$Y_t\left(1 - \beta_2\right) = \beta_1 + I_t + u_t$$

and hence

$$Y_t = \frac{\beta_1}{1-\beta_2} + \frac{1}{1-\beta_2} I_t + \frac{1}{1-\beta_2} u_t. \qquad (1.5)$$

Clearly, from (1.5), Y_t is influenced by u_t. Taking expectations,

$$E\left(Y_t\right) = \frac{\beta_1}{1-\beta_2} + \frac{1}{1-\beta_2} E\left(I_t\right),$$

and therefore

$$\mathrm{Cov}\left(Y_t, u_t\right) = E\left\{\left[Y_t - E\left(Y_t\right)\right]u_t\right\} = E\left\{\left[\frac{1}{1-\beta_2} u_t\right]u_t\right\} = \frac{1}{1-\beta_2} E\left(u_t^2\right) = \frac{\sigma^2}{1-\beta_2} \neq 0.$$

From this derivation it is clear that Y_t and u_t are correlated, and in such a case the application of the OLS method in estimating the parameters will yield biased and inconsistent estimators.

Example 2: Commodity demand model. Consider the commodity demand function representing demand for commodity C:

$$Q_i^c = \alpha_0 + \alpha_1 P_i + \alpha_2 P_i^r + u_i \qquad (1.6)$$

where

- Q_i^c = quantity of commodity C demanded,
- P_i = price of commodity C,
- P_i^r = price of a related commodity,
- u_i = error term,
- α_0, α_1 and α_2 are the parameters.

The model also makes the following assumptions on the error term:

$$E(u_i) = 0, \qquad E(u_i^2) = \sigma^2, \qquad E(u_i u_j) = 0 \ \ (i \neq j), \qquad E\left(u_i P_i^r\right) = 0.$$

Another important dynamic that governs the system is the price equation

$$P_i = \pi_0 + \pi_1 Q_i^c + \pi_2 P_i^r + v_i, \qquad (1.7)$$

with

$$E(v_i) = 0, \qquad E(v_i^2) = \sigma_v^2, \qquad E(v_i v_j) = 0 \ \ (i \neq j), \qquad E\left(v_i P_i^r\right) = 0, \qquad E(u_i v_i) = 0.$$

Consequences of the simultaneous equation. We can now test the assumption of independence between P_i and u_i. Substituting (1.6) into (1.7),

$$P_i = \pi_0 + \pi_1\left(\alpha_0 + \alpha_1 P_i + \alpha_2 P_i^r + u_i\right) + \pi_2 P_i^r + v_i,$$
$$P_i\left(1 - \pi_1\alpha_1\right) = \left(\pi_0 + \pi_1\alpha_0\right) + \left(\pi_1\alpha_2 + \pi_2\right)P_i^r + \left(\pi_1 u_i + v_i\right),$$

so that

$$P_i = \frac{\pi_0 + \pi_1\alpha_0}{1 - \pi_1\alpha_1} + \frac{\pi_1\alpha_2 + \pi_2}{1 - \pi_1\alpha_1}P_i^r + \frac{\pi_1 u_i + v_i}{1 - \pi_1\alpha_1}. \qquad (1.8)$$

Even before any computation is done it is clear that P_i and u_i are dependent. Taking expectations,

$$E\left(P_i\right) = \frac{\pi_0 + \pi_1\alpha_0}{1 - \pi_1\alpha_1} + \frac{\pi_1\alpha_2 + \pi_2}{1 - \pi_1\alpha_1}E\left(P_i^r\right) + \frac{\pi_1 E(u_i) + E(v_i)}{1 - \pi_1\alpha_1}.$$

Calculating the covariance,

$$\mathrm{Cov}\left(P_i, u_i\right) = E\left\{\left[P_i - E\left(P_i\right)\right]u_i\right\} = \frac{\pi_1 E\left(u_i^2\right) + E\left(v_i u_i\right)}{1 - \pi_1\alpha_1} = \frac{\pi_1\sigma^2}{1 - \pi_1\alpha_1} \neq 0.$$

We therefore conclude that P_i and u_i are correlated, in which case the application of OLS will yield biased and inconsistent estimates.

Activity 1.4 — The nature of systems of simultaneous equations

- **Title:** The nature of systems of simultaneous equations.
- **Purpose:** To introduce the concepts of parameter estimation of simultaneous-equation systems.
- **Brief summary of overall task:** Attempt to rewrite the structural equations into reduced forms.
- **Individual contribution:** Consider the supply and demand model

  $$Q_t^D = a_0 + a_1 P_t + a_2 Y_t, \qquad Q_t^S = b_0 + b_1 P_t + b_2 C_t, \qquad Q_t^D = Q_t^S.$$

  i. identify the exogenous and the endogenous variables of the system; (ii) check the identifiability of the system.
- **Spark:** Watch the video on *system of simultaneous equations*.
- **Interaction begins:** Post your answers on discussion forum 1.1; read what your colleagues have posted; in a sentence or two, comment on what two of your colleagues have posted, keeping netiquette in mind.
- **E-moderator interventions:** Focussing group discussion; encouraging lurkers (quiet ones) to contribute; providing feedback/teaching points; closing the discussion.
- **Numbering, pacing and sequencing:** 1.1.
- **Schedule and time:** This activity should take one hour.
- **Next:** Simultaneous equation bias.

1.6 Exercises

1. Consider the following model of supply and demand functions

   $$Q_t^D = a_0 + a_1 P_t + a_2 Y_t, \qquad Q_t^S = b_0 + b_1 P_t + b_2 C_t, \qquad Q_t^D = Q_t^S.$$

   i. Identify the exogenous variables and the endogenous variables of the system.

   ii. Check the identifiability of the system.

1.7 References

Core reading materials

1. Mukras, M. S. (1993). *Elementary Econometrics: Theory, Application and Policy*. East African Educational Publishers Ltd. ISBN 9966-46-557-X.
2. Wooldridge, J. M. (2008). *Introductory Econometrics: A Modern Approach* (4th edition). South-Western. ISBN-10 0324660545, ISBN-13 978-0324660548.

Recommended reading materials

1. Campbell, J. Y., Lo, A. W. and MacKinlay, A. C. (1996). *The Econometrics of Financial Markets* (2nd edition). Princeton University Press. ISBN-10 0691043019, ISBN-13 978-0691043012.
2. Koutsoyiannis, A. (1978). *Theory of Econometrics* (2nd edition). Rowman & Littlefield. ISBN-10 038920563X, ISBN-13 978-0389205630.

3. Greene, W. H. (2011). *Econometric Analysis* (7th edition). Pearson. ISBN-10 0131395386, ISBN-13 978-0131395381.
4. Romer, D. (2011). *Advanced Macroeconomics* (4th edition). McGraw-Hill Education. ISBN-10 0073511374, ISBN-13 978-0073511375.

LESSON TWO: SIMULTANEOUS EQUATION BIAS

2.1 Introduction

In the previous lesson we discussed the assumption of independence between the stochastic term and the regressors, and saw that in most cases it does not hold. As a result, the application of OLS for estimation will yield biased and inconsistent estimates.

2.2 Learning outcomes

By the end of this lesson the learner will be able to:

1. estimate the bias in the estimation of a simultaneous system;
2. demonstrate the asymptotic unbiasedness and the consistency of estimators.

2.3 Equation bias

To demonstrate the consistency and unbiasedness (or otherwise) of the model's estimators, consider an economic model of the form

$$C_t = \beta_1 + \beta_2 Y_t + u_t \qquad (2.1a)$$
$$Y_t = C_t + I_t \qquad (2.1b)$$

where

$$E(u_t) = 0, \qquad E\left(u_t^2\right) = \sigma^2, \qquad E\left(u_t u_{t'}\right) = 0, \qquad E(u_t I_t) = 0.$$

In deviation form — that is, writing

$$c_t = C_t - E\left(C_t\right), \qquad y_t = Y_t - E\left(Y_t\right)$$

— the equation can be written as

$$y_t = \beta_2 c_t + u_t, \qquad (2.1)$$

where

$$E\left(u_t\right) = 0, \qquad E\left(u_t^2\right) = \sigma^2, \qquad E\left(u_t u_{t'}\right) = 0, \qquad E\left(y_t u_t\right) \neq 0.$$

Note that from (2.1b) in deviation form

$$y_t = c_t + i_t, \qquad E\left(y_t u_t\right) = E\left(c_t u_t\right) \neq 0,$$

because the consumption and income deviations both contain the disturbance.

The corresponding **normal equations** are obtained by multiplying equation (2.1) by y_t and summing up:

$$\sum_t y_t^2 = \beta_2 \sum_t y_t c_t + \sum_t y_t u_t. \qquad (2.2)$$

2.3.1 The normal equations and the bias

In the model, y and u are correlated, so $\sum_t y_t u_t$ does not disappear. Dividing equation (2.2) throughout by $\sum_t y_t^2$, and noting that $\sum_t y_t c_t / \sum_t y_t^2$ is the OLS estimator $\widehat{\beta}_2$ applied to (2.1), we obtain

$$\widehat{\beta}_2 = \frac{\sum_t y_t c_t}{\sum_t y_t^2} = \beta_2 + \frac{\sum_t y_t u_t}{\sum_t y_t^2}. \qquad (2.3)$$

Taking expectations of (2.3),

$$E\left(\widehat{\beta}_2\right) = \beta_2 + E\left(\frac{\sum_t y_t u_t}{\sum_t y_t^2}\right) \neq \beta_2,$$

so that the **finite (small) sample bias** is

$$\text{Bias} = E\left(\widehat{\beta}_2\right) - \beta_2 = E\left(\frac{\sum_t y_t u_t}{\sum_t y_t^2}\right) \neq 0.$$

In the case of the asymptotic properties we compute the probability limit of $\widehat{\beta}_2$. Taking probability limits of (2.3),

$$\text{plim}_{n\to\infty}\widehat{\beta}_2 = \beta_2 + \text{plim}_{n\to\infty}\left(\frac{\frac{1}{n}\sum_t y_t u_t}{\frac{1}{n}\sum_t y_t^2}\right) = \beta_2 + \frac{\text{plim}\frac{1}{n}\sum_t y_t u_t}{\text{plim}\frac{1}{n}\sum_t y_t^2}. \qquad (2.4)$$

As n tends to infinity, the sample covariance of Y and u,

$$s_{yu} = \frac{1}{n}\sum_t y_t u_t,$$

tends to the true population covariance of Y and u, which we derived in equation (1.5); and the sample variance of Y,

$$s_y^2 = \frac{1}{n}\sum_t y_t^2,$$

tends to the true population variance of Y, that is $\text{Var}(Y)$.

Equation (2.4) therefore becomes

$$\text{plim}_{n\to\infty}\widehat{\beta}_2 = \beta_2 + \frac{\text{Cov}(Y,u)}{\text{Var}(Y)}. \qquad (2.5)$$

Hence the **asymptotic bias** is given by

$$\text{Asymptotic Bias} = \text{plim}_{n\to\infty}\widehat{\beta}_2 - \beta_2 = \frac{\text{Cov}(Y,u)}{\text{Var}(Y)} = \frac{\sigma^2}{\left(1-\beta_2\right)\text{Var}(Y)} \neq 0,$$

since, from (1.5), $\text{Cov}(Y,u) = \sigma^2/(1-\beta_2)$. The estimator is therefore both biased in small samples and inconsistent: its probability limit differs from the true parameter, so the sample covariance of Y and u and the sample variance of Y do **not** approach values that remove the discrepancy as the sample size increases.

> *[The module's own summary at this point reads: "The sample covariance of Y and u, and the sample variance of Y, as the sample size increases, implies that the estimate approaches the true population values" — a statement about consistency that its own algebra contradicts: because $\text{Cov}(Y,u) \neq 0$, the probability limit in (2.5) is $\beta_2 + \text{Cov}(Y,u)/\text{Var}(Y) \neq \beta_2$. The OLS estimator of a simultaneous equation is inconsistent, which is the whole point of Lessons 5 to 7.]*

Activity 2.4 — Simultaneous equations bias

- **Title:** Bias in simultaneous equation models.
- **Purpose:** To enable the learners to compute the bias for simultaneous-parameter estimators and to differentiate between asymptotic unbiasedness and consistency.
- **Brief summary of overall task:** (i) solve simultaneous equations for parameter estimators; (ii) be able to differentiate between asymptotic unbiasedness and consistency.
- **Individual contribution:** Watch the video *simultaneous equation bias*; read what your colleagues have posted; in a sentence or two, comment on what two of your colleagues have posted, keeping netiquette in mind.
- **Spark:** Watch the video *simultaneous equation bias*.
- **Interaction begins:** Post your answers on discussion forum 1.2.
- **E-moderator interventions:** Focussing group discussion; encouraging lurkers (quiet ones) to contribute; providing feedback/teaching points; closing the discussion.
- **Numbering, pacing and sequencing:** 2.1.
- **Schedule and time:** This activity should take one hour.
- **Next:** Structural and reduced form equations.

2.5 Exercises

Suppose we have the following system of simultaneous equations

$$Y_1 = c_{10} + c_{12}X_2 + c_{11}E_1$$
$$Y_2 = \beta_{20} + \beta_{21}X_1 + E_2,$$

where

$$E\left(E_1\right) = 0, \qquad E\left(E_2\right) = 0, \qquad E\left(E_1^2\right) = \sigma_1^2, \qquad E\left(E_2^2\right) = \sigma_2^2.$$

i. Derive the covariance of X_1 and E.

ii. Approximate the asymptotic bias of X_1 and E.

> *[The system above is reproduced as printed. The symbols of the second question (X_1 and E) are those of the module; the intention is that the covariance of the endogenous regressor with the disturbance, and hence the asymptotic bias of its coefficient, be obtained as in §2.3.]*

2.6 References

Core reading materials

1. Mukras, M. S. (1993). *Elementary Econometrics: Theory, Application and Policy*. East African Educational Publishers Ltd. ISBN 9966-46-557-X.
2. Wooldridge, J. M. (2008). *Introductory Econometrics: A Modern Approach* (4th edition). South-Western. ISBN-10 0324660545, ISBN-13 978-0324660548.

Recommended reading materials

1. Campbell, J. Y., Lo, A. W. and MacKinlay, A. C. (1996). *The Econometrics of Financial Markets* (2nd edition). Princeton University Press.
2. Koutsoyiannis, A. (1978). *Theory of Econometrics* (2nd edition). Rowman & Littlefield.
3. Greene, W. H. (2011). *Econometric Analysis* (7th edition). Pearson.
4. Romer, D. (2011). *Advanced Macroeconomics* (4th edition). McGraw-Hill Education.

LESSON THREE: STRUCTURAL AND REDUCED FORM EQUATIONS

3.1 Introduction

A system of simultaneous equations can be presented either in structural form or in reduced form. In this lesson we demonstrate the distinction between the two forms within a given system, with attention to the notation. Both forms present endogenous variables, denoted by the *Y*'s, and exogenous variables, denoted by the *X*'s. In the structural form the β**'s** are the coefficients attached to the endogenous variables and the γ**'s** are attached to the exogenous variables; the reduced form system will consistently use π**'s** as coefficients of the exogenous variables.

3.2 Learning outcomes

By the end of this lesson the learner will be able to:

1. write a system of simultaneous equations in structural form;
2. write the structural forms into reduced forms.

3.3 Structural equations form

When a system of simultaneous equations is presented in structural form, that presentation is made up of the complete system of equations describing the structure of the relationship of the phenomenon of interest. The equations in the structural model normally express endogenous variables in terms of other endogenous variables, exogenous variables and the stochastic term.

Consider the system given below:

$$C_t = c_1 + c_2 Y_t \qquad (3.1a)$$
$$I_t = i_1 + i_2 Y_{t-1} \qquad (3.1b)$$
$$Y_t = C_t + I_t + G_t \qquad (3.1c)$$
$$T_t = r_1 + r_2 Y_t. \qquad (3.1d)$$

In the model above the endogenous variables are Y, C, I and T, and the exogenous variables are G and Y_{t-1}. Adopting the standard notation and ignoring the intercepts, the system can be written as

$$y_{1t} = \beta_{12} y_{2t} + \gamma_{11} x_{1t} + u_{1t}$$
$$y_{2t} = \beta_{23} y_{3t} + \gamma_{21} x_{2t} + u_{2t}$$
$$y_{3t} = \beta_{31} y_{1t} + \beta_{32} y_{2t} + \gamma_{31} x_{1t} + u_{3t}$$
$$y_{4t} = \beta_{41} y_{1t} + \gamma_{41} x_{3t} + u_{4t},$$

where

$$y_1 = C, \quad y_2 = I, \quad y_3 = Y, \quad y_4 = T; \qquad x_1 = G, \quad x_2 = Y_{t-1}, \quad x_3 = W.$$

> *[The module writes the system in this general β/γ notation (with $x_1 = G$, $x_2 = Y_{t-1}$ and a third predetermined variable), which is the form needed for the order and rank conditions of Lesson 4. The mapping back to (3.1a)–(3.1d) is: $y_1 = C$, $y_2 = I$, $y_3 = Y$, $y_4 = T$.]*

For identification, which we discuss in later lessons, we need to arrange the parameters of the structural model equation by equation. This means re-writing the system with all the variables except the error term on the left-hand side, i.e.

$$y_{1t} - \beta_{12} y_{2t} - \gamma_{11} x_{1t} = u_{1t}$$
$$y_{2t} - \beta_{23} y_{3t} - \gamma_{21} x_{2t} = u_{2t}$$
$$y_{3t} - \beta_{31} y_{1t} - \beta_{32} y_{2t} - \gamma_{31} x_{1t} = u_{3t}$$
$$y_{4t} - \beta_{41} y_{1t} - \gamma_{41} x_{3t} = u_{4t}.$$

The structural parameter equations measure the **direct effects** of the explanatory variables on the dependent variables.

Table of coefficients. The system is then displayed as a table, each row an equation and each column a variable:

Equation	y_1	y_2	y_3	y_4	x_1	x_2	x_3
1	1	$-\beta_{12}$	0	0	$-\gamma_{11}$	0	0
2	0	1	$-\beta_{23}$	0	0	$-\gamma_{21}$	0
3	$-\beta_{31}$	$-\beta_{32}$	1	0	$-\gamma_{31}$	0	0
4	$-\beta_{41}$	0	0	1	0	0	$-\gamma_{41}$

3.4 Reduced form equations

We derive the reduced form equation from the structural equations. In this case the endogenous variables are expressed in terms of the exogenous variables and the error terms. This necessitates the solution of the structural equation system for the *Y*'s in terms of the *X*'s and the error term.

Consider the following structural model of income determination:

$$Y_t = C_t + I_t + G_t \qquad (3.2)$$
$$C_t = c_1 Y_t + u_t \qquad (3.3)$$
$$I_t = i_1 Y_{t-1} + v_t. \qquad (3.4)$$

The endogenous variables of the model are Y, C and I. On the other hand the exogenous variables are G and Y_{t-1}. We can write the reduced model as

$$Y_t = \pi_{11} + \pi_{21} G_t + \pi_{31} Y_{t-1} + w_{1t} \qquad (3.5)$$
$$C_t = \pi_{12} + \pi_{22} G_t + \pi_{32} Y_{t-1} + w_{2t} \qquad (3.6)$$
$$I_t = \pi_{13} + \pi_{23} G_t + \pi_{33} Y_{t-1} + w_{3t}. \qquad (3.7)$$

It is clear from the above case that the structural parameters have been combined in some way to give the reduced forms, expressed strictly with only exogenous variables. The procedure followed is illustrated below.

STEP 1: Substitute (3.3) and (3.4) in (3.2) and solve for Y_t:

$$Y_t = c_1 Y_t + u_t + i_1 Y_{t-1} + v_t + G_t,$$
$$Y_t\left(1 - c_1\right) = G_t + i_1 Y_{t-1} + u_t + v_t,$$
$$Y_t = \frac{1}{1-c_1} G_t + \frac{i_1}{1-c_1} Y_{t-1} + \frac{u_t + v_t}{1-c_1}. \qquad (3.8)$$

Comparing (3.8) with (3.5) gives the reduced form parameters of the income equation:

$$\pi_{21} = \frac{1}{1-c_1}, \qquad \pi_{31} = \frac{i_1}{1-c_1}, \qquad w_{1t} = \frac{u_t + v_t}{1-c_1}.$$

STEP 2: Substitute (3.8) into (3.3) and solve for C_t:

$$C_t = c_1\left[\frac{1}{1-c_1} G_t + \frac{i_1}{1-c_1} Y_{t-1} + \frac{u_t + v_t}{1-c_1}\right] + u_t,$$
$$C_t = \frac{c_1}{1-c_1} G_t + \frac{c_1 i_1}{1-c_1} Y_{t-1} + \frac{c_1\left(u_t + v_t\right)}{1-c_1} + u_t.$$

Comparing with (3.6),

$$\pi_{22} = \frac{c_1}{1-c_1}, \qquad \pi_{32} = \frac{c_1 i_1}{1-c_1}, \qquad w_{2t} = \frac{c_1 u_t + c_1 v_t + \left(1-c_1\right) u_t}{1-c_1} = \frac{u_t + c_1 v_t}{1-c_1}.$$

STEP 3: Substitute (3.8) in (3.4) and solve for I_t:

$$I_t = i_1 Y_{t-1} + v_t,$$

so that, comparing with (3.7),

$$\pi_{23} = 0, \qquad \pi_{33} = i_1, \qquad w_{3t} = v_t.$$

> *[The module states the reduced forms (3.5)–(3.7) and the three steps, but its printed algebra for the coefficients is fragmented. The coefficients above are those that follow from carrying out the three substitutions; the parameter equations in §3.5 are then solved from them, and they reproduce the module's own conclusions about direct and indirect effects.]*

3.5 Direct and indirect effects of parameters

The structural parameters measure the **direct** effect of an explanatory variable on the dependent variable, whereas the reduced form parameters measure the **total** effects of the relevant variable. Consider equations (3.3) and (3.5):

$$C_t = c_1 Y_t + u_t, \qquad Y_t = \frac{1}{1-c_1} G_t + \frac{i_1}{1-c_1} Y_{t-1} + \frac{u_t + v_t}{1-c_1}.$$

We see that $\pi_{31} = \frac{i_1}{1-c_1}$ measures the **total effect** of Y_{t-1} on Y_t: a change in the lagged income works on current income both directly, through the investment function, and indirectly, through consumption. The total effect decomposes as

$$\underbrace{\frac{i_1}{1-c_1}}_{\text{total}} = \underbrace{i_1}_{\text{direct}} + \underbrace{\frac{c_1 i_1}{1-c_1}}_{\text{indirect}}, \qquad \text{i.e.} \quad \pi_{31} = \pi_{33} + \pi_{32} c_1.$$

To determine the indirect effect we decompose the total effect into its two components. Hence

$$\text{TOTAL EFFECT} = \text{DIRECT EFFECT} + \text{INDIRECT EFFECT}.$$

The direct effect of Y_{t-1} on I_t can be easily appreciated: it is i_1, the coefficient of Y_{t-1} in the investment function (3.4).

The indirect effect on I_t can be traced as follows. When Y_{t-1} is added to the investment function (3.4) it increases I_t. When I_t increases, Y_t increases in (3.2), since income is a function of investment; and with an increase in Y_t, consumption C_t increases in (3.3), which in turn raises I_t again through the accelerator term in (3.4). Hence Y_{t-1} has an **indirect** effect on I_t as well, of size $\frac{c_1 i_1}{1-c_1}$.

Activity 3.6 — Structural and reduced equation forms

- **Title:** Structural and reduced equation forms.
- **Purpose:** To enable the learners to re-write structural equation forms into reduced form equations and to explain the implication of the structural and reduced form parameters.
- **Brief summary of overall task:** Distinguish endogenous from predetermined variables in the model; using *y*'s for endogenous variables and *x*'s for predetermined variables, rewrite the system ignoring the intercepts and placing only the error terms on the right of the equal sign; tabulate the coefficients of the system against the corresponding equations and variables; write the structural form sets of simultaneous equations; represent the reduced form equations from the structural ones and interpret the implication of the reduced form parameters.
- **Individual contribution:** In the following Keynesian model of income determination
  $$C_t = a_0 + a_1 Y_t, \qquad I_t = b_0 + b_1 Y_{t-1}, \qquad Y_t = C_t + I_t + G_t, \qquad M_t = m_0 + m_1 Y_t,$$
  carry out the tasks above.
- **Spark:** Watch the video *structural equations and their reduced forms*.
- **Interaction begins:** Post your answers on discussion forum 3.1; read what your colleagues have posted; in a sentence or two, comment on what two of your colleagues have posted, keeping netiquette in mind.
- **E-moderator interventions:** Watch the video *structural equations and their reduced forms*; focussing group discussion; encouraging lurkers (quiet ones) to contribute; providing feedback/teaching points; closing the discussion.
- **Numbering, pacing and sequencing:** 3.1.
- **Schedule and time:** This activity should take two hours.
- **Next:** Problem of identification.

3.6 Exercises

In the following Keynesian model of income determination

$$C_t = a_0 + a_1 Y_t, \qquad I_t = b_0 + b_1 Y_{t-1}, \qquad Y_t = C_t + I_t + G_t, \qquad M_t = m_0 + m_1 Y_t,$$

i. distinguish endogenous from predetermined variables in the model;

ii. using y's for endogenous variables and x's for predetermined variables and the corresponding coefficients, rewrite this system, ignoring the intercepts and placing only the error terms on the right of the equal sign;
iii. tabulate the coefficients of the system against the corresponding equations and variables;
iv. write out the model in reduced form;
v. express the reduced form coefficients in terms of the structural parameters;
vi. show the direct effect, the indirect effects and the total effects of Y_{t-1} on I_t.

3.7 References

See §2.6; the module prints the same core and recommended readings at the end of every lesson.

LESSON FOUR: THE PROBLEM OF IDENTIFICATION

4.1 Introduction

In most econometric problems the issue at hand is always specification, estimation and the evaluation of the statistical reliability of the econometric models which describe a given scenario. Simultaneous equations are treated in the same manner: we specify, estimate and thereafter test the reliability of the models. In this lesson we discuss the problem of identification, to pave the way for the estimation of simultaneous equation systems.

4.2 Learning outcomes

By the end of this lesson the learner will be able to:

1. define the problem of identification and distinguish between exactly identified, over-identified and under-identified equations;
2. discuss the process of model identification following the rules involved;
3. distinguish the order condition from the rank condition of identification.

4.3 Definition of the problem of identification

Identification is the process of establishing whether or not the structural parameters of a simultaneous equation model can be estimated from the parameters of a reduced form model, i.e. the system of simultaneous equations is considered identified when the structural parameters can be estimated from the corresponding reduced form estimates. Identification comes before estimation, once the structural model has been fully specified. The choice of estimation method for the parameters depends on the identification stage.

An equation in a system of equations is said to be **identified** if it is possible to estimate its structural parameters from the reduced form equation system. The equation in such a case will have a unique statistical form, enabling unique estimates of its parameters to be subsequently made from the sample data. If a model is not identified then we do not get unique values of the estimates of the parameters. A system of simultaneous equations is said to be identified if all its equations are identified and it has a unique statistical form: hence there will be no other equation in the system, or algebraic combination of the other equations, which would have the same variables as the equation in question.

An identified equation can be either **exactly identified** or **over-identified**. When the system is exactly identified we can uniquely derive the structural estimates from the reduced form model; in this case the number of parameters to be estimated is equal to the number of independent equations in the system. If it is over-identified, then infinitely many solutions are possible. An equation in a system is said to be **under-identified** if it is not possible to estimate all the structural parameters from the corresponding reduced form. In other words there would be other equation(s), not independent, in the system which can be manipulated to construct another equation containing the same variables as the equation under consideration. Hence an equation is identified if it is independent of any other equation in the system.

Example 4.1. We now discuss the problem of identification by considering the single-commodity market model illustrated below:

$$Q_t^D = \alpha_0 + \alpha_1 P_t + u_{1t} \qquad \text{(demand)}$$
$$Q_t^S = \beta_0 + \beta_1 P_t + u_{2t} \qquad \text{(supply)}$$
$$Q_t^D = Q_t^S \qquad \text{(equilibrium)}$$

where

Q_t^D = quantity demanded, Q_t^S = quantity supplied, P_t = price of the commodity.

We want to find out whether the estimates which we may obtain by using the sample data on demand and price can be identified as estimates of the true demand parameters α_0 and α_1 (and of the supply parameters β_0 and β_1).

We can combine all three equations so as to get one equation. Equating demand and supply,

$$\alpha_0 + \alpha_1 P_t + u_{1t} = \beta_0 + \beta_1 P_t + u_{2t},$$

$$P_t\left(\alpha_1 - \beta_1\right) = \left(\beta_0 - \alpha_0\right) + \left(u_{2t} - u_{1t}\right),$$

$$P_t = \frac{\beta_0 - \alpha_0}{\alpha_1 - \beta_1} + \frac{u_{2t} - u_{1t}}{\alpha_1 - \beta_1}. \qquad (4.1)$$

We therefore have two equations,

$$Q_t = \left(\alpha_0 + \alpha_1 \frac{\beta_0 - \alpha_0}{\alpha_1 - \beta_1}\right) + \alpha_1 \frac{u_{2t} - u_{1t}}{\alpha_1 - \beta_1} + u_{1t} = \pi_1 + v_{1t}, \qquad (4.2)$$

$$Q_t = \left(\beta_0 + \beta_1 \frac{\beta_0 - \alpha_0}{\alpha_1 - \beta_1}\right) + \beta_1 \frac{u_{2t} - u_{1t}}{\alpha_1 - \beta_1} + u_{2t} = \pi_2 + v_{2t}, \qquad (4.3)$$

of the same statistical form and containing the same parameters. However, the first contains the demand parameters (α_0 and α_1) and the second contains the supply parameters (β_0 and β_1). Regressing Q on P with sample data, we cannot be sure whether the estimates we obtain are really α's or β's. That is, the parameters cannot be statistically identified. There is no way of establishing the real supply and demand parameters and the results will be meaningless. This is a case of an **unidentified model**: neither the supply nor the demand function is identified, because there is no way of obtaining the values of the structural parameters from the reduced form equations.

4.4 Rules of identification

In order to establish the identification of an equation in a system of simultaneous equations we need to apply a set of identification rules. There are two distinct approaches used to accomplish this task:

1. the first approach involves the analysis of the **structural form** model;
2. the second approach involves the analysis of the **reduced form** model.

4.4.1 The structural model approach

(i) The order condition for identification

The order condition is a **necessary but not sufficient** condition for identification. This rule involves the comparison of the number of predetermined variables (defined as exogenous and lagged variables) excluded from the equation under discussion with the number of endogenous variables included in the same equation.

Rule. A necessary condition for the identification of a given structural equation in a system of simultaneous equations is that the number of predetermined variables excluded from the given equation is at least as large as the number of endogenous variables included in the equation less one.

For an equation to be identified, the total number of variables excluded from the equation must be at least as great as the number of equations of the system less one.

Let

- G = total number of equations in the model;
- K = total number of variables in the model (exogenous and endogenous);
- M = number of variables, endogenous and exogenous, included in a particular equation.

Then the order condition for identification may be symbolically expressed as

Condition	Conclusion
$K - M = G - 1$	the equation is **exactly identified**
$K - M > G - 1$	the equation is **over-identified**
$K - M < G - 1$	the equation is **under-identified**

Example 4.2. Consider the following system of simultaneous equations:

$$y_1 = 2y_2 + u_1 \qquad (1)$$
$$y_2 = 3y_1 + x_1 + u_2 \qquad (2)$$
$$y_3 = x_2 + u_3 \qquad (3)$$

Check the identifiability of the first equation.

Solution. Total number of variables in the model:

$$K = \underbrace{y_1,\ y_2,\ y_3}_{\text{endogenous}},\ \underbrace{x_1,\ x_2}_{\text{exogenous}} \quad \Rightarrow \quad K = 5.$$

$$G = \text{total number of equations} = 3, \qquad M = \text{variables included in equation (1)} = 2\ \ (y_1,\ y_2).$$

$$K - M = 5 - 2 = 3, \qquad G - 1 = 3 - 1 = 2.$$

Since $K - M = 3 > G - 1 = 2$, equation (1) is **over-identified**.

> *[The module's own figures for this example are $K = 6$, $M = 4$, $K - M = 2$ and $G - 1 = 2$, concluding that the equation is exactly identified. Counting the system as printed — three endogenous variables (y_1, y_2, y_3) and two exogenous (x_1, x_2), with equation (1) containing two variables — gives $K = 5$, $M = 2$, $K - M = 3 > G - 1$. Page 31 of the module then lists the variables of the "first equation" as y_1, y_2, y_3, x_3, which belong to no equation of this system, so the printed figures appear to have been carried over from a different model. The rule itself is stated correctly above and is applied in Examples 4.3 and 4.4.]*

Example 4.3. Determine whether or not the following two equations satisfy the order condition:

$$y_1 = 2y_2 + x_1 + x_2 + u_1 \qquad (1)$$
$$y_2 = 3y_1 + u_2. \qquad (2)$$

Solution.

Equation 1. Total variables in the model:

$$K = \underbrace{y_1,\ y_2}_{\text{endogenous}}\ ,\ \underbrace{x_1,\ x_2}_{\text{exogenous}} \quad \Rightarrow \quad K = 4.$$

$$G = 2, \qquad M = \text{variables in equation (1)} = 4\ \ (y_1,\ y_2,\ x_1,\ x_2).$$

$$K - M = 4 - 4 = 0, \qquad G - 1 = 2 - 1 = 1.$$

Since $K - M = 0 < G - 1 = 1$, **equation one is under-identified**.

Equation 2.

$$K = 4, \qquad G = 2, \qquad M = \text{variables in equation (2)} = 3\ \ (y_1,\ y_2 \text{ and } u_2),$$

$$K - M = 4 - 3 = 1, \qquad G - 1 = 2 - 1 = 1.$$

Since $K - M = G - 1$, equation (2) satisfies the order condition and hence **is identified**.

> *[In equation 2 the module counts $M = 3$, i.e. the two endogenous variables plus the disturbance; the disturbance is not one of the K variables, but the conclusion $K - M = G - 1$ is unaffected if M is taken as 2 in the notation of equation (1), where the count was 4 for the same reason. The two counts are therefore consistent with each other.]*

Another approach is to construct the table of coefficients (parameters) of both the endogenous and the predetermined variables; thereafter the predetermined and endogenous variables of any one equation can be counted and compared.

Example 4.4. Establish and determine whether the first three equations of the Keynesian model given below satisfy the necessary condition for identification. Consider the system:

$$C_t = \alpha_0 + \alpha_1 Y_t + \alpha_2 T_t + u_{1t} \qquad (1) \quad \text{consumption function}$$
$$I_t = \beta_0 + \beta_1 Y_{t-1} + \beta_2 Y_t + u_{2t} \qquad (2) \quad \text{investment function}$$

$$T_t = \gamma_0 + \gamma_1 Y_t + u_{3t} \qquad (3) \quad \text{tax function}$$
$$Y_t = C_t + I_t + G_t \qquad (4) \quad \text{identity.}$$

In the model above the endogenous variables are C, I, T and Y, and the predetermined variables are G and Y_{t-1}.

Table of coefficients

Equation	C	T	Y	I	G	Y_{t-1}
(1)	1	$-\alpha_2$	$-\alpha_1$	0	0	0
(2)	0	0	$-\beta_2$	1	0	$-\beta_1$
(3)	0	1	$-\gamma_1$	0	0	0
(4)	-1	0	1	-1	-1	0

We now check the identifiability of each equation.

Equation (1):

$$K = 6, \quad G = 4, \quad M = 3 \ (C, T, Y);$$
$$K - M = 3, \quad G - 1 = 3 \quad \Rightarrow \quad K - M = G - 1.$$

Equation (1) is **identified** (exactly).

Equation (2):

$$K = 6, \quad G = 4, \quad M = 3 \ (I, Y, Y_{t-1});$$
$$K - M = 3, \quad G - 1 = 3 \quad \Rightarrow \quad K - M = G - 1.$$

Equation (2) is **identified**.

> *[The module prints $M = 2$ for equation (2), giving $K - M = 4 > G - 1 = 3$ and calling the equation over-identified. Equation (2) as printed contains three variables — I_t, Y_t and Y_{t-1} — so $M = 3$ and the equation is exactly identified. The rank condition of Example 4.5, which the module finds to be satisfied for equation (2), is consistent with either classification.]*

Equation (3):

$$K = 6, \quad G = 4, \quad M = 2 \ (T, Y);$$
$$K - M = 4, \quad G - 1 = 3 \quad \Rightarrow \quad K - M > G - 1.$$

Equation (3) is **identified** (over-identified).

> *[The module prints $M = 1$ and $K - M = 5$ for equation (3). As printed, the tax function contains T_t and Y_t, so $M = 2$ and $K - M = 4$; the conclusion — over-identified — is the same.]*

Equation (4) is not an econometric equation, since the variables do not have coefficients attached to them; it is an identity and is not identified in the same sense.

(ii) The rank condition for identification

This condition is called the rank condition because it entails the computation of the rank of the matrix of parameters of the variables **excluded** from the particular equation being identified. The **rank of a matrix** is the order of the largest non-zero determinant which can be formed from the matrix of parameters of the excluded variables.

The rank condition is the **sufficient** condition and can be stated in two ways.

Rule. In a system of G equations, any particular equation is identified if and only if it is possible to construct at least one non-zero determinant of order $G - 1$ from the coefficients of the parameters excluded from that particular equation but contained in the other equations of the model.

Rule. An equation in a system of G simultaneous equations will be identified if the rank of the matrix of parameters of all the variables excluded (endogenous and predetermined) from the equation is equal to $G - 1$. Denoting this matrix by Q, the rule means that

$$\text{Rank}(Q) = G - 1.$$

Example 4.5. Establish and determine whether the first three equations of the Keynesian model of Example 4.4 satisfy the sufficient condition for identification.

Solution. The table of all parameters of the model is the one given in Example 4.4. We now generate the table of parameters of the variables **excluded** from each equation in turn.

Table of parameters of variables excluded from equation (1) — the excluded variables are I, G and Y_{t-1}:

Equation	I	G	Y_{t-1}
(2)	1	0	$-\beta_1$
(3)	0	0	0
(4)	-1	-1	0

The relevant matrix of order $G-1=3$ is

$$Q_1=\begin{pmatrix} 1 & 0 & -\beta_1 \\ 0 & 0 & 0 \\ -1 & -1 & 0 \end{pmatrix}.$$

Expanding along the second row, $\det(Q_1)=0$, hence $\text{rank}(Q_1)=2<G-1=3$. We conclude that **the rank condition is not satisfied and therefore equation (1) is not identified** — even though its order condition was satisfied, which is exactly why the order condition is only necessary.

Table of parameters of variables excluded from equation (2) — the excluded variables are C, T and G:

Equation	C	T	G
(1)	1	$-\alpha_2$	0
(3)	0	1	0
(4)	-1	0	-1

The matrix of the table is

$$Q_2=\begin{pmatrix} 1 & -\alpha_2 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & -1 \end{pmatrix}.$$

Several matrices of order $G-1=3$ can be formed from this table; two of them are

$$Q_{21}=\begin{pmatrix} 1 & -\alpha_2 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & -1 \end{pmatrix}, \qquad Q_{22}=\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & -1 & 0 \end{pmatrix},$$

with

$$\det\left(Q_{21}\right)=1\cdot 1\cdot(-1)=-1\neq 0 \quad \Rightarrow \quad \text{rank}\left(Q_{21}\right)=3=G-1.$$

Hence the rank of Q_2 is $G-1=3$, the rank condition is satisfied, and **equation (2) is identified**. We need prove it for only one of the matrices which can be generated that the determinant is not equal to zero.

Table of parameters of variables excluded from equation (3) — the excluded variables are C, I and G:

Equation	C	I	G
(1)	1	0	0
(2)	0	1	0
(4)	-1	-1	-1

The matrix is

$$Q_3=\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & -1 \end{pmatrix},$$

and several matrices of order $G-1=3$ can be generated; three of them are

$$Q_{31}=\begin{pmatrix}1&0&0\\0&1&0\\-1&-1&-1\end{pmatrix},\qquad Q_{32}=\begin{pmatrix}1&0&0\\0&1&0\\-1&-1&-1\end{pmatrix},\qquad Q_{33}=\begin{pmatrix}1&0&0\\0&0&0\\-1&-1&-1\end{pmatrix},$$

with

$$\det\left(Q_{31}\right)=1\cdot 1\cdot(-1)=-1\neq 0\quad\Rightarrow\quad \operatorname{rank}\left(Q_{31}\right)=3=G-1,$$

$$\det\left(Q_{33}\right)=0\quad\Rightarrow\quad \operatorname{rank}\left(Q_{33}\right)=2.$$

Since at least one determinant of order $G-1=3$ is non-zero, the rank condition is satisfied and **equation (3) is identified**.

> *[The matrices printed in the module for equations (2) and (3) are partly obscured — the page shows rows such as "0 1 0" and "1 0 0 0" without their columns. The matrices above are the sub-matrices of the coefficient table of Example 4.4; they give the module's own determinants and its conclusions (equation (1) unidentified; equations (2) and (3) identified).]*

4.4.2 The reduced model approach

In the case of the reduced form model there are also two conditions for identification: the **order condition**, which is necessary but not sufficient, and the **rank condition**, which is necessary and sufficient.

The order condition is the same as in the case of the structural model, but the rank condition is slightly different: it is based on selected parameters of the reduced form model.

The rank condition. A structural equation with G endogenous variables is identified if it is possible to construct at least one **non-zero** determinant of order $G-1$ from the reduced form parameters corresponding to the included endogenous and the excluded predetermined variables of the structural equation.

Example 4.6. Consider the following system of simultaneous equations. By means of the reduced form approach, investigate the state of identification of each equation.

$$y_1=2y_2+x_1+u_1\qquad(1)$$
$$y_2=y_3+x_2+u_2\qquad(2)$$
$$y_3=2y_1+x_3+x_2+u_3.\qquad(3)$$

We first convert the above simultaneous system into the structural equation model. Since equations (1), (2) and (3) have no intercepts and are written in terms of the y's and x's, the system is already in structural form.

Substituting equation (1) in (3):

$$y_3=2\left(2y_2+x_1+u_1\right)+x_3+x_2+u_3=4y_2+2x_1+x_2+x_3+2u_1+u_3.\qquad(4)$$

Substituting (4) in (2):

$$y_2=\left(4y_2+2x_1+x_2+x_3+2u_1+u_3\right)+x_2+u_2,$$
$$y_2\left(1-4\right)=2x_1+2x_2+x_3+2u_1+u_2+u_3,$$
$$y_2=-\tfrac{2}{3}x_1-\tfrac{2}{3}x_2-\tfrac{1}{3}x_3-\tfrac{2}{3}u_1-\tfrac{1}{3}u_2-\tfrac{1}{3}u_3.\qquad(5)$$

Substituting (4) and (5) in (3) — or substituting (5) in (1) — gives

$$y_1=2\left(-\tfrac{2}{3}x_1-\tfrac{2}{3}x_2-\tfrac{1}{3}x_3\right)+x_1+(\text{errors})=-\tfrac{1}{3}x_1-\tfrac{4}{3}x_2-\tfrac{2}{3}x_3+(\text{errors}),$$

$$y_3=2y_1+x_3+x_2+u_3=-\tfrac{2}{3}x_1-\tfrac{8}{3}x_2-\tfrac{4}{3}x_3+x_3+x_2+(\text{errors})=-\tfrac{2}{3}x_1-\tfrac{5}{3}x_2-\tfrac{1}{3}x_3+(\text{errors}).\qquad(6)$$

Dropping the error terms we obtain the reduced form equations

$$y_1=\pi_{11}x_1+\pi_{12}x_2+\pi_{13}x_3,\qquad y_2=\pi_{21}x_1+\pi_{22}x_2+\pi_{23}x_3,\qquad y_3=\pi_{31}x_1+\pi_{32}x_2+\pi_{33}x_3,$$

with $\pi_{11}=-\tfrac{1}{3},\pi_{12}=-\tfrac{4}{3},\pi_{13}=-\tfrac{2}{3};\ \pi_{21}=-\tfrac{2}{3},\pi_{22}=-\tfrac{2}{3},\pi_{23}=-\tfrac{1}{3};$ and $\pi_{31}=-\tfrac{2}{3},\pi_{32}=-\tfrac{5}{3},\pi_{33}=-\tfrac{1}{3}.$

We investigate the state of identification of equation (1) using the order and the rank conditions.

(a) Equation (1) — order condition.

$$K=6\ (y_1,\ y_2,\ y_3,\ x_1,\ x_2,\ x_3),\qquad M=4\ (y_1,\ y_2,\ x_1\text{ and the disturbance}),\qquad K-M=2;$$
$$G=3\quad\Rightarrow\quad G-1=2.$$

Since $K-M=G-1$, the order condition is satisfied and equation (1) is exactly identified.

(b) Equation (1) — rank condition. In equation (1) the endogenous variables included are y_1 and y_2, and the excluded predetermined variable is x_2 (and x_3). We need to construct two tables from the reduced form model:

1. the table of **all** reduced form parameters;
2. the table of reduced form parameters corresponding to the endogenous variables included in, and the predetermined variables excluded from, the equation of interest.

Table 1. All reduced form parameters

	x_1	x_2	x_3
y_1	1	-2	-3
y_2	2	-2	-2
y_3	1	-2	-4

Table 2. Reduced form parameters of the included endogenous and excluded predetermined variables

	x_2	x_3
y_1	-2	-3
y_2	-2	-2

We now proceed to check whether there exists a non-zero determinant of order $G - 1 = 1$. The non-zero entries which can be constructed from Table 2 are four; taking the two entries of the x_2 column and of the x_3 column,

$$\det(-2) = -2 \neq 0 \qquad \text{and} \qquad \det(-2) = -2 \neq 0.$$

We conclude that the rank condition is satisfied, and since the order condition had also been satisfied, equation one is **fully identified**.

> *[The reduced-form substitution (4)–(6) is worked out above because the module's printed algebra for it is fragmentary; the table of reduced form parameters is reproduced as the module prints it, and it does yield the module's conclusion (equation (1) fully identified). The module then sets the identification of equations (2) and (3) as the exercise below. Note also that the module's later discussion of the reduced-form rank condition speaks of constructing a "zero determinant", while the rule it states — and uses — requires a **non-zero** determinant of order $G - 1$.]*

Exercise. Investigate the state of identification of equations (2) and (3) of Example 4.6.

Activity 4.4 — Problem of identification in simultaneous equations

- **Title:** Problem of identification.
- **Purpose:** To enable the learners to comprehend the order and rank conditions for the identification of simultaneous equations.
- **Brief summary of overall task:** (i) write the structural form sets of simultaneous equations; (ii) represent the reduced form equations from the structural ones and interpret the implication of the reduced form parameters; (iii) read Chapter 17, *Econometrics: Simultaneous Equations Models*.
- **Individual contribution:** Watch the video *structural equations and their reduced forms*; read what your colleagues have posted; in a sentence or two, comment on what two of your colleagues have posted, keeping netiquette in mind.
- **Spark:** Watch the video *structural equations and their reduced forms*.
- **Interaction begins:** Post your answers on discussion forum 4.1.
- **E-moderator interventions:** Focussing group discussion; encouraging lurkers (quiet ones) to contribute; providing feedback/teaching points; closing the discussion.
- **Numbering, pacing and sequencing:** 4.1.
- **Schedule and time:** This activity should take two hours.
- **Next:** Simultaneous equation estimation.

4.5 Exercises

1. Consider the following model of supply and demand functions:

$$Q_t^D = a_0 + a_1 P_t + a_2 Y_t$$
$$Q_t^S = b_0 + b_1 P_t + b_2 C_t$$
$$Q_t^D = Q_t^S \qquad \text{(at equilibrium)},$$

where D = quantity demanded, S = quantity supplied, P = price of the commodity, Y = income, and t = time trend (in the demand function it stands for 'tastes' and in the supply function for 'technology').

   i. Examine the identification condition for each equation by the order condition for identification.

   ii. If any of the functions is over-identified, what restrictions would be imposed for the function to be identified?

   iii. Construct the corresponding reduced form model and determine whether or not the rank condition is satisfied.

2. Consider the following system of simultaneous equations:

$$C_t = a_0 + a_1 Y_t$$
$$I_t = b_0 + b_1 Y_{t-1} + b_2 Y_t$$
$$Y_t = C_t + I_t + G_t.$$

   i. Examine the identification condition for each equation by the order condition for identification.

   ii. If any of the functions is over-identified, what restrictions would be imposed for the function to be identified?

   iii. Construct the corresponding reduced form model and determine whether or not the rank condition is satisfied.

> *[The two systems above are the ones printed at the head and foot of the module's page 43; they are brought together here as questions 1 and 2, since the page sets out each of them with the same three sub-questions.]*

4.6 References

See §2.6; the module prints the same core and recommended readings at the end of every lesson.

LESSON FIVE: SIMULTANEOUS EQUATION ESTIMATION — INDIRECT LEAST SQUARES

5.1 Introduction

We found that when there is double causation in our structural model, ordinary least squares estimators are inconsistent and biased owing to the correlation between the residuals and the regressor(s). **Theil** and **Basmann** worked out this problem and independently came up with methods of estimating these parameters consistently.

The various methods available for estimating simultaneous equation models can be classified under two main groups:

1. **single-equation methods**, also called the *limited information methods*;
2. **systems methods**, also called the *full information methods*.

The single-equation methods were designed to estimate a single structural equation separately, with only reference to the rest of the system. Information from the rest of the system mainly pertains to the identifiability of the particular equation; this is why this set of methods is referred to as "limited information methods". The most popular single-equation methods are ordinary least squares (OLS), **indirect least squares (ILS)**, the **instrumental variable method (IV)** and the **two-stage least squares method (2SLS)**.

The systems methods, on the other hand, are used to estimate all the equations simultaneously, using information from all the equations — parameter restrictions of all equations as well as the variances and covariances of the residuals. The systems methods include three-stage least squares (3SLS), the full-information least generalized variance (FI/LGV) and the full-information maximum likelihood (FI/ML) methods.

5.2 Learning outcomes

By the end of this lesson the learner will be able to:

1. discuss the indirect least squares estimation method;
2. discuss the theoretical properties of estimators obtained using the indirect least squares estimation method.

5.2.1 Identification and choice of estimation method

Once an equation in a system of simultaneous equations is identified, the econometrician needs to decide on the appropriate estimation method. In general identification determines the choice of econometric method for estimation:

State of identification	Feasible estimation
under-identified	the equation cannot be estimated, so no estimation technique arises
exactly identified	indirect least squares (ILS) is the most appropriate
over-identified	two-stage least squares (2SLS) is the method popularly used

5.3 The indirect least squares method (ILS)

The ILS method, sometimes called the **reduced form method**, is a single-equation method used in estimating exactly identified equations. In the simultaneous equation system the explanatory variables include endogenous variables; the ILS adopts the strategy of transforming the structural model such that all the endogenous variables are regressed on the predetermined variables in the system. This strategy eliminates the correlations that occur between some endogenous explanatory variables and the stochastic terms in the structural model.

5.3.1 Assumptions of the ILS method

Assumption 1. The structural equation must be **exactly identified**. If the structural equation is unidentified (under-identified) then the structural parameters cannot be estimated, since the number of estimating equations would be less than the number of unknowns to be estimated.

Assumption 2. The stochastic terms in the reduced form equations must satisfy the usual assumptions of ordinary least squares. This is because the ILS method applies OLS to the reduced form equations to estimate the parameters of those equations.

1. *The assumption of randomness of* u_i. The term u_i is assumed to be random: it can take either a positive or a negative value.
2. *The assumption of zero mean of* u_i. $E(u_i) = 0$.
3. *The assumption of homoscedasticity.* The variance of u_i is constant, that is to say u_i is homoscedastic: $E\left(u_i^2\right) = \sigma^2$.
4. *The assumption of normality.* The stochastic term is assumed to be normally distributed with mean zero and constant variance, that is $u_i \sim N\left(0, \sigma^2\right)$.
5. *The assumption of zero covariance of* X_i *and* u_i. The variable X_i and the stochastic term are non-correlated and therefore have zero covariance, $\text{Cov}\left(X_i, u_i\right) = 0$. This is because X_i is a fixed constant.
6. *The assumption of non-autocorrelation of* u_i *and* u_j. The covariance of the stochastic term for period i and that for period j is equal to zero, that is they are uncorrelated with each other: $E\left(u_i u_j\right) = 0, i \neq j$.

Assumption 3. There should not be strong collinearity among the exogenous variables.

5.3.2 The process of ILS

The process of ILS is outlined below.

Step 1. Construct the **reduced form model** from the structural model. In the reduced form the explanatory variables are made up of predetermined variables alone and are therefore not correlated with the error term, hence the situation of double causation is avoided.

Step 2. Write the reduced form model in **two forms**. In the initial form, done in Step 1, the parameters are a mixture of the structural model parameters; in the second form the parameters of the model are denoted by π's.

Step 3. Form the **system of parameter equations** from these two sets of equations.

Step 4. Apply OLS to the reduced model expressed in the second form, so as to obtain the estimates of the π's.

Step 5. By substituting the estimates for the π's in the system of parameter equations, solve for the **structural parameters**. The values obtained will be unique if the structural model is exactly identified.

5.4 Examples

Example 1. Consider the following Keynesian model:

$$C_t = \beta_1 + \beta_2 Y_t + u_t \qquad (1)$$

$$Y_t = C_t + I_t. \qquad (2)$$

Use the ILS method to derive the structural estimates of the model.

SOLUTION

STEP 1: Check whether the structural equations are identified. Use the rank condition, which is necessary and sufficient.

$$G - 1 = 2 - 1 = 1.$$

We need to form at least one non-zero determinant of order 1 from the variables excluded from equation (2). Equation (2) contains Y_t, C_t and I_t; the variables excluded from it are only those appearing in equation (1). Forming the matrix of excluded variables,

$$A_Q = \begin{pmatrix} -1 \end{pmatrix}, \qquad \det\left(A_Q\right) = -1 \neq 0.$$

Hence, using the rank condition, **equation (2) is exactly identified**; and since the system is identified, equation (1) is too. The ILS method is therefore the appropriate estimation method.

> *[The module's coefficient table for this model is printed as two rows — "1, 1, –1, –1" and "2, 1, 0, –1" against the variables — with the second equation's excluded variable giving the* 1×1 *matrix* $A_Q = (-1)$ *and* $\det(A_Q) = -1$*, exactly as above.]*

STEP 2: Construct the corresponding reduced form model. Substituting (1) in (2),

$$Y_t = \beta_1 + \beta_2 Y_t + u_t + I_t, \qquad Y_t = \frac{\beta_1}{1-\beta_2} + \frac{1}{1-\beta_2} I_t + \frac{1}{1-\beta_2} u_t,$$

so the reduced forms are

$$Y_t = \pi_{11} + \pi_{21} I_t + w_{1t} \qquad (3)$$
$$C_t = \pi_{12} + \pi_{22} I_t + w_{2t}, \qquad (4)$$

where, from (3),

$$\pi_{11} = \frac{\beta_1}{1-\beta_2}, \qquad \pi_{21} = \frac{1}{1-\beta_2}, \qquad w_{1t} = \frac{u_t}{1-\beta_2},$$

and, substituting (3) into (1),

$$C_t = \beta_1 + \beta_2\left(\pi_{11} + \pi_{21} I_t + w_{1t}\right) + u_t = \left(\beta_1 + \beta_2 \pi_{11}\right) + \beta_2 \pi_{21} I_t + \left(\beta_2 w_{1t} + u_t\right),$$

so that

$$\pi_{12} = \beta_1 + \beta_2 \pi_{11} = \frac{\beta_1}{1-\beta_2}, \qquad \pi_{22} = \beta_2 \pi_{21} = \frac{\beta_2}{1-\beta_2}, \qquad w_{2t} = \beta_2 w_{1t} + u_t.$$

STEP 3: Form the system of parameter equations from the two forms of the reduced model:

$$\pi_{21} = \frac{1}{1-\beta_2} \quad \Rightarrow \quad 1-\beta_2 = \frac{1}{\pi_{21}} \quad \Rightarrow \quad \beta_2 = 1 - \frac{1}{\pi_{21}} = \frac{\pi_{21}-1}{\pi_{21}},$$

$$\pi_{22} = \frac{\beta_2}{1-\beta_2} = \beta_2 \pi_{21} \quad \Rightarrow \quad \beta_2 = \frac{\pi_{22}}{\pi_{21}},$$

$$\pi_{11} = \pi_{12} = \frac{\beta_1}{1-\beta_2} = \beta_1 \pi_{21} \quad \Rightarrow \quad \beta_1 = \frac{\pi_{11}}{\pi_{21}} = \frac{\pi_{12}}{\pi_{21}}.$$

STEP 4: Apply OLS to the reduced forms so as to estimate the π's:

$$\widehat{\pi}_{11} = \bar{Y} - \widehat{\pi}_{21}\bar{I}, \qquad \widehat{\pi}_{21} = \frac{\sum_t \left(I_t - \bar{I}\right)\left(Y_t - \bar{Y}\right)}{\sum_t \left(I_t - \bar{I}\right)^2},$$

$$\widehat{\pi}_{12} = \bar{C} - \widehat{\pi}_{22}\bar{I}, \qquad \widehat{\pi}_{22} = \frac{\sum_t \left(I_t - \bar{I}\right)\left(C_t - \bar{C}\right)}{\sum_t \left(I_t - \bar{I}\right)^2}.$$

STEP 5: Substitute the estimates of the π's into the system of parameter equations to obtain the estimates of the structural parameters:

$$\widehat{\beta}_2 = \frac{\widehat{\pi}_{21} - 1}{\widehat{\pi}_{21}} = \frac{\widehat{\pi}_{22}}{\widehat{\pi}_{21}}, \qquad \widehat{\beta}_1 = \frac{\widehat{\pi}_{11}}{\widehat{\pi}_{21}} = \frac{\widehat{\pi}_{12}}{\widehat{\pi}_{21}}.$$

The two expressions for $\widehat{\beta}_2$ agree only if the exactly-identified restrictions hold in the sample; this coincidence of two estimates of the same parameter is the signature of an exactly identified equation.

> *[The module states Steps 1–5 and prints the reduced forms and parameter equations in its own fragmented layout. The algebra above carries the steps through, and reproduces the module's own identification finding (*$\det A_Q = -1$*; equation (2) exactly identified).]*

5.4.1 Properties of ILS estimates

The ILS method is used to estimate the structural parameters via the reduced form estimates. If the assumptions of the ILS method are fulfilled, then:

(i) The reduced form estimators will have the following properties:

a. linear;

b. unbiased;

c. least variance.

(ii) The structural estimators will have the following properties:

a. biased for small samples;

b. consistent — that is, the sample bias tends to zero as the sample size increases and the distribution of the estimator collapses on the true value.

Example 2. Consider the following system of simultaneous equations:

$$y_1 = \alpha_1 y_2 + \alpha_2 x_1 + u_1 \qquad (1)$$
$$y_2 = \beta_1 y_1 + \beta_2 x_2 + \beta_3 x_3 + u_2. \qquad (2)$$

i. Determine the identification condition of each of the equations;

ii. derive the corresponding reduced form model;

iii. estimate the structural parameters of the equations of the system.

Solution.

(i) Identification. We first construct the table of coefficients of the system of equations.

Equation	y_1	y_2	x_1	x_2	x_3
(1)	1	$-\alpha_1$	$-\alpha_2$	0	0
(2)	$-\beta_1$	1	0	$-\beta_2$	$-\beta_3$

Order condition. $G = 2, K = 5$.

Equation 1: $M = 3$ $(y_1,\ y_2,\ x_1)$, so $K - M = 2$ and $G - 1 = 1$; since $K - M > G - 1$ the order condition is satisfied.

Equation 2: $M = 4$ (y_1, y_2, x_2, x_3), so $K - M = 1 = G - 1$; the order condition is satisfied.

Rank condition, using the structural model approach.

Equation 1: the variables excluded from equation (1) are x_2 and x_3, so matrices of order $G - 1 = 1$ formed from their coefficients in equation (2) are

$$A_1 = \begin{pmatrix} -\beta_2 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} -\beta_3 \end{pmatrix}, \qquad A_3 = \begin{pmatrix} -\beta_2 & -\beta_3 \end{pmatrix}.$$

Since $\det(A_1) = -\beta_2 \neq 0$ and $\det(A_2) = -\beta_3 \neq 0$, the rank condition is satisfied. **Conclusion:** because the order and the rank conditions are satisfied, equation 1 is **over-identified** (two distinct non-zero determinants of order 1 can be formed).

Equation 2: the variable excluded from equation (2) is x_1, so the matrix of excluded variables is

$$A_1 = \begin{pmatrix} -\alpha_2 \end{pmatrix}, \qquad \det(A_1) = -\alpha_2 \neq 0.$$

Only one non-zero determinant of order 1 can be formed, and the rank condition is satisfied. **Conclusion:** equation (2) is **exactly identified**.

> *[The module prints "M = 4" for equation 2 and concludes "Equation 2 is exactly identified", and "M = 3, K – M > G – 1" for equation 1 with the conclusion "over-identified"; the counting above reproduces both. Because equation (1) is over-identified and equation (2) exactly identified, the system as a whole is over-identified, which is why the module's second example foreshadows 2SLS (Lesson 7) for equation (1).]*

(ii) Reduced form. Substituting (2) in (1) to get (3):

$$y_1 = \alpha_1\left(\beta_1 y_1 + \beta_2 x_2 + \beta_3 x_3 + u_2\right) + \alpha_2 x_1 + u_1,$$
$$y_1\left(1 - \alpha_1\beta_1\right) = \alpha_2 x_1 + \alpha_1\beta_2 x_2 + \alpha_1\beta_3 x_3 + \left(u_1 + \alpha_1 u_2\right),$$

$$y_1 = \frac{\alpha_2}{1-\alpha_1\beta_1}x_1 + \frac{\alpha_1\beta_2}{1-\alpha_1\beta_1}x_2 + \frac{\alpha_1\beta_3}{1-\alpha_1\beta_1}x_3 + \frac{u_1+\alpha_1u_2}{1-\alpha_1\beta_1}. \qquad (3)$$

Substituting (3) in (2) we get (4):

$$y_2 = \beta_1\left(\pi_{11}x_1 + \pi_{12}x_2 + \pi_{13}x_3 + w_1\right) + \beta_2x_2 + \beta_3x_3 + u_2,$$

$$y_2 = \beta_1\pi_{11}x_1 + \left(\beta_1\pi_{12} + \beta_2\right)x_2 + \left(\beta_1\pi_{13} + \beta_3\right)x_3 + \left(\beta_1w_1 + u_2\right), \qquad (4)$$

where $\pi_{11} = \frac{\alpha_2}{1-\alpha_1\beta_1}$, $\pi_{12} = \frac{\alpha_1\beta_2}{1-\alpha_1\beta_1}$, $\pi_{13} = \frac{\alpha_1\beta_3}{1-\alpha_1\beta_1}$.

(iii) Structural estimates. The reduced equations (3) and (4) carry the coefficients π_{ij}, and the system of parameter equations they imply is

$$\pi_{11} = \frac{\alpha_2}{1-\alpha_1\beta_1}, \qquad \pi_{12} = \frac{\alpha_1\beta_2}{1-\alpha_1\beta_1}, \qquad \pi_{13} = \frac{\alpha_1\beta_3}{1-\alpha_1\beta_1}, \qquad (5)$$

$$\pi_{21} = \beta_1\pi_{11}, \qquad \pi_{22} = \beta_1\pi_{12} + \beta_2, \qquad \pi_{23} = \beta_1\pi_{13} + \beta_3. \qquad (6)$$

Apply OLS to the reduced equations to get the estimates of the reduced-form coefficients, $\widehat{\pi}_{11}, \ldots, \widehat{\pi}_{23}$. Substituting these estimates into the system of parameter equations (5) and (6), we obtain

$$\widehat{\pi}_{11} = \frac{\widehat{\alpha}_2}{1-\widehat{\alpha}_1\widehat{\beta}_1}, \qquad \widehat{\pi}_{12} = \frac{\widehat{\alpha}_1\widehat{\beta}_2}{1-\widehat{\alpha}_1\widehat{\beta}_1}, \qquad \widehat{\pi}_{13} = \frac{\widehat{\alpha}_1\widehat{\beta}_3}{1-\widehat{\alpha}_1\widehat{\beta}_1}, \qquad (7),\ (8),\ (9)$$

$$\widehat{\pi}_{21} = \widehat{\beta}_1\widehat{\pi}_{11}, \qquad \widehat{\pi}_{22} = \widehat{\beta}_1\widehat{\pi}_{12} + \widehat{\beta}_2, \qquad \widehat{\pi}_{23} = \widehat{\beta}_1\widehat{\pi}_{13} + \widehat{\beta}_3. \qquad (10),\ (11),\ (12)$$

Solve equations (7) to (12) to get the estimates of the structural model:

- Divide (10) by (7): since $\widehat{\pi}_{21} = \widehat{\beta}_1\widehat{\pi}_{11}$, we obtain $\widehat{\beta}_1 = \widehat{\pi}_{21}/\widehat{\pi}_{11}$.
- Multiply (8) by $\widehat{\beta}_1$ and subtract from (11): since $\widehat{\pi}_{22} = \widehat{\beta}_1\widehat{\pi}_{12} + \widehat{\beta}_2$, we obtain $\widehat{\beta}_2 = \widehat{\pi}_{22} - \widehat{\beta}_1\widehat{\pi}_{12}$.
- Substitute into (9): since $\widehat{\pi}_{23} = \widehat{\beta}_1\widehat{\pi}_{13} + \widehat{\beta}_3$, we obtain $\widehat{\beta}_3 = \widehat{\pi}_{23} - \widehat{\beta}_1\widehat{\pi}_{13}$.

With $\widehat{\beta}_1$, $\widehat{\beta}_2$ and $\widehat{\beta}_3$ now known, equation (7) gives $\widehat{\alpha}_1$ and equation (8) or (9) then gives $\widehat{\alpha}_2$:

$$\widehat{\alpha}_1 = \frac{\widehat{\pi}_{12}}{\widehat{\beta}_2 + \widehat{\beta}_1\widehat{\pi}_{12}} = \frac{\widehat{\pi}_{12}}{\widehat{\pi}_{22}}, \qquad \widehat{\alpha}_2 = \widehat{\pi}_{11}\left(1 - \widehat{\alpha}_1\widehat{\beta}_1\right).$$

> *[The module's solution stops at the three instructions just carried out — "Divide (10) with (7)", "Multiply (8) by ___ and subtract from (11)", "Substitute ___ in (8) — then substitute in (9)" — with the symbols detached from their slots. The three identities above are the operations those instructions perform, and the last line gives $\widehat{\alpha}_1$ and $\widehat{\alpha}_2$ from equations (7) and (8). Note also that equation (1) of this example is over-identified, so ILS does not in fact deliver a unique structural estimate for it: the parameter equations are not all independent, and the example serves to show why over-identified equations require 2SLS — which is the subject of Lesson 6. The module presents the exercise without remarking on this.]*

5.6 E-tivity 5 — Indirect least squares estimation method

Activity 5.6 — The indirect least squares estimation method

- **Title:** The indirect least squares estimation method.
- **Purpose:** To enable the learners to estimate parameters of an exactly identified function in a system of simultaneous equations.
- **Brief summary of overall task:** The indirect least squares estimation method, and the two-stage least squares method (2SLS).
- **Individual contribution:** Watch the video *The Indirect Least Squares Method*; (i) estimate the structural parameters using the ILS method; (ii) discuss some theoretical properties of the estimators obtained.
- **Interaction begins:** Post your answers on discussion forum 5.1.
- **E-moderator interventions:** Focussing group discussion; encouraging lurkers (quiet ones) to contribute; providing feedback/teaching points; closing the discussion.
- **Numbering, pacing and sequencing:** 5.1.
- **Schedule and time:** This activity should take two hours.
- **Next:** The two-stage least squares method (2SLS).

5.7 Exercises

Consider the following system of simultaneous equations:

$$Y_1 = \alpha_0 + \alpha_1 Y_2 + \alpha_2 I + u_1, \qquad (1)$$
$$Y_2 = \beta_0 + \beta_1 Y_1 + \beta_2 M + u_2, \qquad (2)$$

where Y_1 and Y_2 are the two endogenous variables of the system — Y_1 = income, Y_2 = money supply — I = investment expenditure and M is the remaining predetermined variable.

1. Determine the identification condition of each of the equations.
2. Derive the corresponding reduced-form model.
3. Estimate the structural parameters of the equations of the system.

> *[The exercise's two equations are printed as the runs "IYY" / "122101" and "YY" / "21102", with the variable key on the following page reading "*___* = Income, *___* = money supply, I = investment expenditure". The system above is that reading: income and money supply endogenous, investment expenditure and one further predetermined variable as regressors. The module's own numerals are not recoverable in order, so the equation numbers and the coefficient subscripts are consecutive; the identification analysis the exercise asks for is unaffected.]*

5.8 References

Core reading materials

1. Mukras, M. S. (1993). *Elementary Econometrics: Theory, Application and Policy*. East African Educational Publishers Ltd. ISBN 9966-46-557-X.
2. Wooldridge, J. M. (2008). *Introductory Econometrics: A Modern Approach* (4th edition). South-Western. ISBN-10 0324660545, ISBN-13 978-0324660548.

Recommended reading materials

1. Campbell, J. Y., Lo, A. W., MacKinlay, A. C. and Lo, A. Y. (1996). *The Econometrics of Financial Markets* (2nd edition). Princeton University Press. ISBN-10 0691043019, ISBN-13 978-0691043012.
2. Koutsoyiannis, A. (1978). *Theory of Econometrics* (2nd edition). Rowman & Littlefield Publishers. ISBN-10 038920563X, ISBN-13 978-0389205630.

3. Greene, W. H. (2011). *Econometric Analysis* (7th edition). Pearson. ISBN-10 0131395386, ISBN-13 978-0131395381.
4. Romer, D. (2011). *Advanced Macroeconomics* (4th edition). McGraw-Hill Education. ISBN-10 0073511374, ISBN-13 978-0073511375.

LESSON SIX: THE TWO-STAGE LEAST SQUARES METHOD (2SLS)

6.1 Introduction

This method is applicable to single **over-identified** equations. As given in the previous lesson, detailed discussions and the method are outlined in (Mukras, 1993) pages 492 up to 495. Similar discussions are also given in (Koutsoyiannis, 1978) and (Wooldridge, 2008). The method can also be applied to exactly identified equations, in which case the estimates will be identical to those of ILS.

> *[The module's introduction names the references exactly as reproduced above. Its spelling of Koutsoyiannis varies between lessons ("Koutsoyiannia" on this page); the correct spelling is used throughout this edition.]*

6.2 Learning outcomes

By the end of this lesson the learner will be able to:

6.2.1 discuss the assumptions on which the 2SLS method is based and outline the procedure of 2SLS;

6.2.2 apply the 2SLS method on over-identified equations of a system of simultaneous equations;

6.2.3 discuss the theoretical properties of estimators obtained for the 2SLS method.

6.2.1 Assumptions of the 2SLS method

The 2SLS method is based on the assumptions given below. We consider a system of over-identified structural equations,

$$y_1 = \beta_{10} + \beta_{11}y_2 + \beta_{12}x_1 + \beta_{13}x_3 + u_1, \qquad (6.1)$$
$$y_2 = \beta_{20} + \beta_{21}y_1 + \beta_{22}x_2 + \beta_{23}x_3 + \beta_{24}x_4 + u_2. \qquad (6.2)$$

We now construct:

(a) the reduced form equation for the endogenous variables, which are regressed on all the predetermined variables of the system:

$$y_1 = \pi_{11}x_1 + \pi_{12}x_2 + \pi_{13}x_3 + \pi_{14}x_4 + v_1,$$
$$y_2 = \pi_{21}x_1 + \pi_{22}x_2 + \pi_{23}x_3 + \pi_{24}x_4 + v_2;$$

(b) and write out the estimated equations:

$$\widehat{y}_1 = \widehat{\pi}_{11}x_1 + \widehat{\pi}_{12}x_2 + \widehat{\pi}_{13}x_3 + \widehat{\pi}_{14}x_4, \qquad (6.3)$$
$$\widehat{y}_2 = \widehat{\pi}_{21}x_1 + \widehat{\pi}_{22}x_2 + \widehat{\pi}_{23}x_3 + \widehat{\pi}_{24}x_4. \qquad (6.4)$$

> *[The subscripts of the model on this page are printed as loose numerals ("13131122111", "24233221212"), so the two structural equations cannot be read off the page symbol by symbol. They have been re-set with consecutive subscripts, preserving what the page fixes: equation (6.1) is the equation of y_1 and contains y_2 as a regressor, and equation (6.2) is the equation of y_2; four predetermined variables x_1, …, x_4 appear in the system, which is what the reduced forms on the same page regress on.]*

ASSUMPTION 1

It is assumed that the structural equations at issue are identified — more precisely, that the equation being estimated is over-identified, which is the case the method is designed for.

ASSUMPTION 2

The stochastic terms in the original structural-form equations must satisfy the usual assumptions of ordinary least squares. This is because the second stage of the method applies OLS to the transformed structural equation. The six requirements are:

(i) The assumption of randomness of u. The term u is assumed to be random. It can assume either a positive or a negative value.

(ii) The assumption of zero mean of u.

$$E\left(u_i\right) = 0.$$

(iii) The assumption of homoscedasticity. The variance of u is constant, that is, u is homoscedastic:

$$E\left(u_i^2\right) = \sigma^2.$$

(iv) The assumption of normality. The stochastic term is assumed to be normally distributed with mean zero and constant variance σ^2:

$$u \sim N\left(0, \sigma^2\right).$$

(v) The assumption of zero covariance of X and u. The variable X and the stochastic term u are non-correlated and therefore have zero covariance:

$$\text{Cov}\left(X_i, u_i\right) = 0.$$

This is because X is a fixed constant.

(vi) The assumption of non-autocorrelation of u_i and u_j. This means that the covariance of the stochastic term for period i, u_i, and that for period j, u_j, is equal to zero — that is, they are uncorrelated with each other:

$$E\left(u_i u_j\right) = 0, \qquad i \neq j.$$

ASSUMPTION 3

The stochastic terms in the **reduced-form** equations must satisfy the usual assumptions of ordinary least squares. This is because the first stage of the method applies OLS to the reduced-form equations in order to obtain the estimated values $\widehat{y}_1$ and $\widehat{y}_2$ that replace the endogenous regressors. The six requirements are the same as in Assumption 2, with v in place of u:

(i) randomness of v;

(ii) zero mean, $E\left(v_i\right) = 0$;

(iii) homoscedasticity, $E\left(v_i^2\right) = \sigma_v^2$;

(iv) normality, $v \sim N\left(0, \sigma_v^2\right)$;

(v) zero covariance of X and v, $\text{Cov}\left(X_i, v_i\right) = 0$, since X is a fixed constant;

(vi) non-autocorrelation, $E\left(v_i v_j\right) = 0$ for $i \neq j$.

> *[Assumption 3 in the module carries the sentence "This is because in the ILS method applies the OLS to the reduced form equations", which is copied from the previous lesson. Here it has been corrected to describe the first stage of 2SLS, which is the reason the assumption is needed.]*

ASSUMPTION 4

It is assumed that the econometrician has included all the important predetermined variables in the specification of the system.

ASSUMPTION 5

The explanatory variables are not strongly correlated with one another — that is, the system is free of severe multicollinearity.

ASSUMPTION 6

The sample size should be large enough to permit the investigator to obtain significant estimates of the reduced-form coefficients. At the very least, the number of observations should be more than the number of predetermined variables in the structural system.

6.2.2 Outline of the 2SLS procedure

STEP 1. Identify the endogenous variable appearing also as an explanatory variable in the equation under consideration. Also identify all the predetermined variables in the system. Take, for example, equation (6.2). In this case the relevant variables are $y_1, y_2, x_1, x_2, x_3, x_4$.

STEP 2. Obtain the reduced-form equation by regressing the endogenous variable on all the predetermined variables in the system. We get

$$y_1 = \pi_{11}x_1 + \pi_{12}x_2 + \pi_{13}x_3 + \pi_{14}x_4 + v_1,$$
$$y_2 = \pi_{21}x_1 + \pi_{22}x_2 + \pi_{23}x_3 + \pi_{24}x_4 + v_2.$$

STEP 3. By the OLS method, estimate the reduced-form coefficients so as to obtain $\widehat{\pi}_{ij}$, and thus

$$\widehat{y}_1 = \widehat{\pi}_{11}x_1 + \widehat{\pi}_{12}x_2 + \widehat{\pi}_{13}x_3 + \widehat{\pi}_{14}x_4,$$
$$\widehat{y}_2 = \widehat{\pi}_{21}x_1 + \widehat{\pi}_{22}x_2 + \widehat{\pi}_{23}x_3 + \widehat{\pi}_{24}x_4.$$

STEP 4. Replace the endogenous variable which was playing the role of an explanatory variable — and hence causing bias in the estimation — by its estimated value:

$$y_1 = \beta_{10} + \beta_{11}\widehat{y}_2 + \beta_{12}x_1 + \beta_{13}x_3 + u_1^*,$$

where $u_1^* = u_1 + \beta_{11}\widehat{v}_2$ is an error term and $\widehat{v}_2 = y_2 - \widehat{y}_2$.

STEP 5. The transformed equation now contains only predetermined variables and a composite error term, so we apply the OLS method a **second time** — to this transformed equation — to get the estimates of the structural coefficients. Hence the reason why the method is referred to as a 2SLS method.

> *[The module's Step 4 reads "where ___ an error term and" with the definition detached. The composite error $u_1^* = u_1 + \beta_{11}\hat{v}_2$ is the one implied by the substitution made in that step. The module's Step 5 opens "The term ___ is an intercept in the transformed equation"; the term that matters for the method is the composite error just defined, and that is what is stated here.]*

6.2.3 Application of the two-stage least squares method

Consider the following system of two simultaneous equations:

$$y_1 = \beta_{11}y_2 + \beta_{12}x_1 + u_1, \qquad (1)$$
$$y_2 = \beta_{21}y_1 + \beta_{22}x_2 + \beta_{23}x_3 + v_2. \qquad (2)$$

1. Determine the state of identification of each of the equations in the system.
2. State which estimation method would be appropriate for each of the two equations.
3. Explain how you would proceed to estimate the first equation in the system.

SOLUTION. We first construct the table of coefficients of the system of equations. The system contains the two endogenous variables y_1 and y_2 and the three predetermined variables x_1, x_2, x_3, so that

$$G = 2, \qquad K = 5.$$

Order condition (equation 1)

The equation contains $M = 3$ variables — y_1, y_2, x_1 — so

$$K - M = 5 - 3 = 2 \qquad \text{and} \qquad G - 1 = 1, \qquad \text{therefore} \qquad K - M > G - 1.$$

The order condition is satisfied. According to the order condition, equation 1 is **over-identified**.

Rank condition (equation 1)

With $G-1=2-1=1$, where G is the number of equations in the system, the determinants of order $G-1$ of the variables excluded from the equation are formed from the columns of x_2 and x_3 — the two predetermined variables that the system contains but equation 1 omits:

$$A_1=\begin{pmatrix}\beta_{22}\end{pmatrix},\qquad A_2=\begin{pmatrix}\beta_{23}\end{pmatrix}.$$

Since these determinants are non-zero, the rank condition is satisfied. Since the number of non-zero determinants is more than one, the equation is over-identified.

Conclusion. Equation 1 is over-identified, since both the order condition (inequality maintained) and the rank condition are satisfied.

Order condition (equation 2)

The equation contains $M=4$ variables — y_2, y_1, x_2, x_3 — so

$$K-M=5-4=1 \quad \text{and} \quad G-1=1, \quad \text{hence} \quad K-M=G-1.$$

The order condition is satisfied. According to the order condition, equation (2) is **exactly identified**.

Rank condition (equation 2)

With $G-1=1$, the determinant of order $G-1$ of the variable excluded from equation 2 is formed from the column of x_1:

$$A_1=\begin{pmatrix}\beta_{12}\end{pmatrix}.$$

Since this determinant is non-zero, the rank condition is satisfied. Since there is only one non-zero determinant, equation (2) is exactly identified.

Which method. Since equation 1 is over-identified, we can use the **2SLS** method for its estimation. Equation 2 is exactly identified, so the **ILS** method is used for it.

Estimating equation 1 by 2SLS

The equation to be estimated is

$$y_1=\beta_{10}+\beta_{11}y_2+\beta_{12}x_1+u_1. \qquad (6.5)$$

STEP 1. Regress y_2 on all the predetermined variables of the system, obtaining the reduced-form equation, and estimate it by OLS:

$$y_2=\pi_{21}x_1+\pi_{22}x_2+\pi_{23}x_3+v_2, \qquad \widehat{y}_2=\widehat{\pi}_{21}x_1+\widehat{\pi}_{22}x_2+\widehat{\pi}_{23}x_3.$$

STEP 2. Replace y_2 in (6.5) by $\widehat{y}_2$ and apply the OLS method so as to obtain the structural estimates:

$$y_1=\beta_{10}+\beta_{11}\widehat{y}_2+\beta_{12}x_1+u_1^*, \qquad u_1^*=u_1+\beta_{11}\widehat{v}_2.$$

The relevant normal equations are

$$\begin{aligned}\sum y_1&=n\widehat{\beta}_{10}+\widehat{\beta}_{11}\sum\widehat{y}_2+\widehat{\beta}_{12}\sum x_1,\\ \sum\widehat{y}_2y_1&=\widehat{\beta}_{10}\sum\widehat{y}_2+\widehat{\beta}_{11}\sum\widehat{y}_2^2+\widehat{\beta}_{12}\sum\widehat{y}_2x_1,\\ \sum x_1y_1&=\widehat{\beta}_{10}\sum x_1+\widehat{\beta}_{11}\sum x_1\widehat{y}_2+\widehat{\beta}_{12}\sum x_1^2.\end{aligned}$$

Solving these three equations gives $\widehat{\beta}_{10}, \widehat{\beta}_{11}$ and $\widehat{\beta}_{12}$.

For equation (2). Let us apply the ILS method, because the equation is exactly identified: derive its reduced form, estimate the reduced-form coefficients by OLS, and solve the parameter equations for the structural coefficients.

[The module prints the identification analysis of this example in full, but the normal equations of the second stage survive only as detached fragments ("$\sum x_1 y_1$", "$\widehat{\beta}_{11} \sum x_1 \hat{y}_2$", "$\widehat{\beta}_{12} \sum x_1^2$", "$\sum \hat{y}_2^2$", ...). The three equations above are the standard normal equations of the second-stage regression, assembled from those fragments; the coefficients they solve for are the ones the module names.]

6.2.4 Properties of 2SLS estimators

1. For small samples the 2SLS estimators are biased, but for large samples they are asymptotically unbiased — that is, they are **consistent**.
2. Their variance tends to zero as the sample increases to infinity; that is, the distribution collapses on the true value of the parameter.
3. They have smaller asymptotic variance within the class of consistent estimators.

6.2.5 Advantages of 2SLS estimators

1. The 2SLS method can be applied to exactly identified equations, although it is designed for over-identified equations.
2. Unlike ILS, which will give multiple parameter estimates for over-identified equations, 2SLS gives **only one** estimate for each parameter.
3. For estimating econometric models involving many equations, 2SLS is economical, in that it can be applied to individual equations in the system without directly taking into account any other equation in the system.

6.3 E-tivity 6 — The Two-Stage Least Squares Estimation Method (2SLS)

Activity 6.3 — The two-stage least squares estimation method

- **Title:** The two-stage least squares estimation method.
- **Purpose:** To enable the learners to estimate parameters of an over-identified function in a system of simultaneous equations.
- **Brief summary of overall task:** The two-stage least squares estimation method, and the instrumental variable method.
- **Individual contribution:** Watch the video *The Two-Stage Least Squares Method*.
- **Interaction begins:** Post your answers on discussion forum 6.1.
- **E-moderator interventions:** Focussing group discussion; encouraging lurkers (quiet ones) to contribute; providing feedback/teaching points; closing the discussion.
- **Numbering, pacing and sequencing:** 7.1.
- **Schedule and time:** This activity should take two hours.
- **Next:** The instrumental variable method (IV).

[The activity plan is printed as an interleaved two-column table, and its cells carry the module's own numbering "7.1" — the numbering of the next lesson, another sign of the copy-forward by which these pages were assembled. The field values above are assigned by meaning.]

6.4 Exercise

1. Consider the following system of simultaneous equations:

$$y_1 = \beta_{11} y_2 + \beta_{12} x_1 + u_1,$$
$$y_2 = \beta_{21} y_1 + \beta_{22} x_2 + \beta_{23} x_3 + v_2.$$

i. Determine the identification condition of each of the equations.

ii. Derive the corresponding reduced-form model.

iii. Estimate the structural parameters of the equations of the system.

2. The money market of an economy is defined with the following model:

$$\text{Demand: } M_d = \beta_{10} + \beta_{11}Y + \beta_{12}r + \beta_{13}P + u_1,$$
$$\text{Supply: } M_s = \beta_{20} + \beta_{21}Y + \beta_{22}Y_{t-1} + \beta_{23}M_{t-1} + u_2,$$

where M_d is the demand for money, M_s the supply of money, Y is income, r the rate of interest and P the price level.

i. Determine the identification state of each of the equations in the system.

ii. State the methods you would use to estimate the functions.

iii. After finding out that Y_{t-1} and M_{t-1} are statistically insignificant, the supply function is re-defined excluding Y_{t-1} and M_{t-1}. What is the identification state of each of the equations in this case?

iv. Which estimation methods would you use in this case?

> *[The two systems in this exercise are printed with their symbols in a single run at the foot of the page ("β_{13}", "P", "r", "Y", "M_d", "M_s", "M_{t-1}", "Y_{t-1}" ...). The models above are assembled from those symbols together with the variable list the page itself gives (Y = income, r = rate of interest, P = price, "Demand and Supply of money"), and the money-market specification is the one the exercise's own part (iii) presupposes.]*

6.5 References

Core reading materials

1. Mukras, M. S. (1993). *Elementary Econometrics: Theory, Application and Policy*. East African Educational Publishers Ltd. ISBN 9966-46-557-X.
2. Wooldridge, J. M. (2008). *Introductory Econometrics: A Modern Approach* (4th edition). South-Western. ISBN-10 0324660545, ISBN-13 978-0324660548.

Recommended reading materials

1. Campbell, J. Y., Lo, A. W., MacKinlay, A. C. and Lo, A. Y. (1996). *The Econometrics of Financial Markets* (2nd edition). Princeton University Press. ISBN-10 0691043019, ISBN-13 978-0691043012.
2. Koutsoyiannis, A. (1978). *Theory of Econometrics* (2nd edition). Rowman & Littlefield Publishers. ISBN-10 038920563X, ISBN-13 978-0389205630.
3. Greene, W. H. (2011). *Econometric Analysis* (7th edition). Pearson. ISBN-10 0131395386, ISBN-13 978-0131395381.
4. Romer, D. (2011). *Advanced Macroeconomics* (4th edition). McGraw-Hill Education. ISBN-10 0073511374, ISBN-13 978-0073511375.

LESSON SEVEN: THE INSTRUMENTAL VARIABLE METHOD (IV)

7.1 The IV method

This method is applicable to **over-identified** equations. As given in the previous lesson, detailed discussions and the method are outlined in (Mukras, 1993) pages 485 up to 491. Similar discussions are also given in (Koutsoyiannis, 1978) and (Wooldridge, 2008).

7.2 Learning outcomes

By the end of this lesson the learner will be able to:

7.2.1 apply the IV method on over-identified equations of a system of simultaneous equations;

7.2.2 discuss the theoretical properties of estimators obtained for the IV method.

7.2.1 Assumptions of the instrumental variable method (IV)

The method of instrumental variables is based on the following assumptions.

ASSUMPTION 1: It is assumed that the structural equations at issue are identified.

ASSUMPTION 2: It is assumed that the econometrician has knowledge of some exogenous variables in the system which may be used as instrumental variables in the equations under consideration.

ASSUMPTION 3: The chosen instrument must possess the following properties:

1. It must be **strongly correlated** with the endogenous explanatory variable which is correlated with the error term in the structural equation. This is the correlation that the instrument intends to reduce or eliminate.
2. It should be **uncorrelated with the exogenous variables** already appearing as explanatory variables in the structural equation under consideration. This requirement is aimed at the reduction or elimination of multicollinearity.
3. It should be uncorrelated with other instrumental variables to be used in the structural equation, where more than one instrument is needed.
4. It should be uncorrelated with the exogenous variable already appearing as an explanatory variable in the structural equation. We should note that these exogenous variables are usually used as their own instruments in the estimation process.

ASSUMPTION 4: The new stochastic term — the transformed term obtained after transforming the original structural equation — must satisfy the usual assumptions of OLS which we defined earlier.

ASSUMPTION 5: The exogenous variables (the instrumental variables) should not be strongly multicollinear.

> *[The module numbers this fifth item "v) ASSUMPTION 4" and then "vi) ASSUMPTION 5", continuing a list that Assumption 3 did not begin. The five assumptions are numbered 1 to 5 here; no content is changed.]*

7.2.2 The instrumental variable method

This method aims at removing as much as possible the dependence existing between the endogenous explanatory variables and the stochastic term. This dependence, which gives rise to the violation of one of the assumptions of OLS, is removed by **transforming the structural equation** in such a manner as to make the application of the OLS method appropriate.

Outline of the method

STEP 1. Ensure that the assumptions of the instrumental variable method, already presented above, are satisfied.

STEP 2. Select an appropriate instrumental variable which will be used for transforming the structural equation under consideration. We need to note two things:

1. The number of instrumental variables needed for the transformation of a particular structural equation is **equal to the number of endogenous variables** appearing within the set of explanatory variables of that equation.
2. If the equation at issue contains exogenous variables in the set of explanatory variables, these will be used as instruments **for themselves**.

STEP 3. Transform the structural equation by multiplying the equation by each one of the instrumental variables, taking account of the fact that the exogenous variables in the equation must also be used as instruments for themselves. This will give us as many equations as there are parameters.

STEP 4. Sum the transformed equations over all sample observations.

STEP 5. Solve these equations so as to obtain parameter estimates for the structural equations.

7.2.3 Example 7.1

Consider the following system of two simultaneous equations:

$$y_1 = \beta_1 y_2 + u_1, \qquad (7.1)$$
$$y_2 = \gamma_1 z_1 + \gamma_2 z_2 + v_2. \qquad (7.2)$$

Using the IV method, derive the estimate of β_1 in the first equation of the system.

SOLUTION.

(i) The structural equation has an endogenous variable y_2 as an explanatory variable. This definitely gives rise to simultaneous-equation bias; and it is clear, since there are two instruments available (z_1 and z_2) for one endogenous regressor, that the equation is **over-identified**.

(ii) We choose z_1 as the instrumental variable for transforming this equation. As can be seen in equation (7.2), z_1 is correlated with y_2; also z_1 is truly exogenous and hence uncorrelated with the stochastic term.

Using the procedure stepwise, we proceed as follows.

STEP 1. Rewrite the equation in structural form and in deviation form:

$$y_1 = \beta_1 y_2 + u_1.$$

STEP 2. Transform the equation by multiplying it by the instrumental variable z_1:

$$z_1 y_1 = \beta_1 z_1 y_2 + z_1 u_1.$$

STEP 3. Sum the equation over all sample observations:

$$\sum z_1 y_1 = \beta_1 \sum z_1 y_2 + \sum z_1 u_1.$$

Note that z_1 and the error term are uncorrelated, meaning that $E\left(z_1 u_1\right) = 0$; thus, by assuming that the last term is equal to zero, we have the estimator

$$\widehat{\beta}_1 = \frac{\sum z_1 y_1}{\sum z_1 y_2}.$$

> *[The module writes the system of this example with its subscripts detached ("zy", "zy", "β_1", "γ_1", "γ_2", "z_1", "z_2"). Equations (7.1) and (7.2) above preserve the structure the solution presupposes: y_2 is regressed on two exogenous variables z_1 and z_2, one of which is used as the instrument for y_2 in the first equation, and the other of which remains available — which is what makes the first equation over-identified.]*

Substitution route. The same estimate can be reached by substitution. Substitute the estimate of y_2 in equation (7.1), then substitute the resulting equation in equation (7.2) to get a reduced equation,

$$y_1 = \pi_{11} z_1 + \pi_{12} z_2 + w_1,$$

where

$$\pi_{11} = \beta_1\gamma_1, \qquad \pi_{12} = \beta_1\gamma_2, \qquad w_1 = \beta_1 v_2 + u_1.$$

The coefficients of the reduced equations can be obtained using the OLS method, and the structural coefficient recovered as $\widehat{\beta}_1 = \widehat{\pi}_{11}/\widehat{\gamma}_1 = \widehat{\pi}_{12}/\widehat{\gamma}_2$ — the two routes agreeing only because the equation is exactly determined by each instrument separately.

7.2.4 Example 7.2

Consider the following system of two simultaneous equations, written in deviation form:

$$y_1 = \beta_1 y_2 + \beta_2 z_1 + u_1, \qquad (7.3)$$
$$y_2 = \gamma_1 z_1 + \gamma_2 z_2 + \gamma_3 z_3 + v_2. \qquad (7.4)$$

Estimate the parameters of both equations by the IV approach.

Solution — equation (7.3).

(i) This equation is over-identified: it contains one endogenous regressor, y_2, and the system supplies three exogenous variables z_1, z_2, z_3, of which z_1 already appears in (7.3) as its own instrument, leaving z_2 and z_3 as instruments for y_2 — more than the one required.

(ii) This equation has one variable, y_2, which is endogenous in equation (7.4) but appears as an explanatory variable in (7.3); hence it is correlated with the error term in equation (7.3). We will eliminate this correlation by transforming equation (7.3) using z_2 or z_3, which are highly correlated with y_2 and are purely exogenous, as instruments.

(iii) We select z_2 as our instrument. We also use z_1 and z_3 as their own instruments — z_1 because it appears in (7.3), and z_3 because, being exogenous and omitted from (7.3), it is available.

(iv) We assume that the sum of the products of the instruments and the error term in equation (7.3) is zero, and hence drop those terms from the equations formed when we transform equation (7.3) by multiplying it by each instrumental variable.

(v) Multiply equation (7.3) by the three instruments z_1, z_2 and z_3, and assume

$$\sum z_1 u_1 = \sum z_2 u_1 = \sum z_3 u_1 = 0.$$

This gives the three normal equations

$$\sum z_1 y_1 = \beta_1 \sum z_1 y_2 + \beta_2 \sum z_1^2, \qquad (7.5)$$
$$\sum z_2 y_1 = \beta_1 \sum z_2 y_2 + \beta_2 \sum z_2 z_1, \qquad (7.6)$$
$$\sum z_3 y_1 = \beta_1 \sum z_3 y_2 + \beta_2 \sum z_3 z_1. \qquad (7.7)$$

The three normal equations are the same in number as the number of unknowns, β_1 and β_2 (together with the intercept where the equation is written in levels). Hence the parameters can be estimated consistently using OLS, giving $\left(\widehat{\beta}_1, \widehat{\beta}_2\right)$.

EXERCISE. Estimate the parameters of equation (7.4) using the IV method.

> *[The module prints the three normal equations as one run of symbols ("$\sum z_3 z_2 y_1$", "$\sum z_3 y_2$", "$\sum z_1^2$", "$\sum z_2 z_1$" …). The equations above are those runs read against the instruments the solution names. The module's sentence "the three normal equations are the same number as the number of three unknowns" counts three unknowns; the equation as printed in deviation form has two slopes, so the third is the intercept, and the note is kept as the module words it.]*

7.3 E-tivity 7 — The Instrumental Variable Method (IV)

Activity 7.3 — The instrumental variable method

- **Title:** The instrumental variable estimation method.
- **Purpose:** To enable the learners to estimate parameters of an over-identified function in a system of simultaneous equations.
- **Brief summary of overall task:** The instrumental variable estimation method.
- **Individual contribution:** Watch the video *The Instrumental Variable Techniques*.
- **Interaction begins:** Post your answers on discussion forum 6.1.
- **E-moderator interventions:** Focussing group discussion; encouraging lurkers (quiet ones) to contribute; providing feedback/teaching points; closing the discussion.
- **Numbering, pacing and sequencing:** 6.1.
- **Schedule and time:** This activity should take two hours.
- **Next:** END OF MODULE.

> *[This activity is printed as "6.4 E-tivity" — the numbering of the previous lesson — although it belongs to Lesson 7; its forum is numbered 6.1 and its "Next" field reads "END OF MODULE." It is numbered 7.3 in this edition, with the module's own numerals kept in the field values.]*

7.4 Exercise

1. Consider the following system of two simultaneous equations written in deviation form:

$$y_1 = \beta_1 y_2 + \beta_2 z_1 + u_1,$$
$$y_2 = \gamma_1 z_1 + \gamma_2 z_2 + \gamma_3 z_3 + v_2.$$

i. Estimate the parameters of the system using the IV method.

ii. What are their theoretical properties?

2. Consider the following system of simultaneous equations:

$$y_1 = \beta_{11} y_2 + \beta_{12} x_1 + u_1,$$
$$y_2 = \beta_{21} y_1 + \beta_{22} x_2 + \beta_{23} x_3 + v_2.$$

i. Determine the identification condition of each of the equations.

ii. Derive the corresponding reduced-form model.

iii. Estimate the structural parameters of the equations of the system.

7.5 References

Core reading materials

1. Mukras, M. S. (1993). *Elementary Econometrics: Theory, Application and Policy*. East African Educational Publishers Ltd. ISBN 9966-46-557-X.
2. Wooldridge, J. M. (2008). *Introductory Econometrics: A Modern Approach* (4th edition). South-Western. ISBN-10 0324660545, ISBN-13 978-0324660548.

Recommended reading materials

1. Campbell, J. Y., Lo, A. W., MacKinlay, A. C. and Lo, A. Y. (1996). *The Econometrics of Financial Markets* (2nd edition). Princeton University Press. ISBN-10 0691043019, ISBN-13 978-0691043012.
2. Koutsoyiannis, A. (1978). *Theory of Econometrics* (2nd edition). Rowman & Littlefield Publishers. ISBN-10 038920563X, ISBN-13 978-0389205630.

3. Greene, W. H. (2011). *Econometric Analysis* (7th edition). Pearson. ISBN-10 0131395386, ISBN-13 978-0131395381.
4. Romer, D. (2011). *Advanced Macroeconomics* (4th edition). McGraw-Hill Education. ISBN-10 0073511374, ISBN-13 978-0073511375.

END OF MODULE.

APPENDIX A — THE STATE OF THE SOURCE FILE

The download `SST 404_ Econometrics II - Comprehensive Lesson Notes - Studocu.html` contains **72 pages, and all 72 carry text**: 3,156 text fragments, every one of them placed on a page by the file's own layout rules. The three lessons that follow identification — ILS, 2SLS and IV — are all present in full, with their e-tivities, exercises and references.

File pages	Content
1	Title page: Kenyatta University, Digital School of Virtual and Open Learning, SST 404 Econometrics II
2–5	Table of contents (Lessons 1–7 and their sections)
6	Course description: purpose, outcomes, content, mode of delivery, assessment, core reading
7	Recommended reading materials
8–14	**Lesson 1** — system of simultaneous equations
15–19	**Lesson 2** — simultaneous equation bias
19–26	**Lesson 3** — structural and reduced form equations
27–44	**Lesson 4** — the problem of identification
45–54	**Lesson 5** — indirect least squares (ILS)
55–65	**Lesson 6** — the two-stage least squares method (2SLS)
66–72	**Lesson 7** — the instrumental variable method (IV), ending "END OF MODULE"

Two earlier versions of this download ended at page 51, in the middle of Example 2 of Lesson 5, with pages 52–72 present but empty. The current file completes that example and carries Lessons 6 and 7 as well, so this edition now covers the whole module. Nothing is missing except the answers to the exercises: the module sets them and supplies no answer key.

APPENDIX B — WHAT WAS BROKEN, AND WHAT WAS REPAIRED

B.1 Three streams of text, laid on top of one another

Each page carried, superimposed:

1. **the lesson prose**, in the body column;
2. **the two-column iLearn activity plan** — *Title, Purpose, Brief summary of overall task, Individual contribution, Spark, Interaction begins, E-moderator interventions, Numbering and pacing, Schedule and time, Next* — whose cells interleaved into a single run; and
3. **the mathematics**.

Read straight out of the file, an activity arrived like this:

> "Spark E-moderator Interaction Post your answers on discussion forum 1.1 Focussing group discussion interventions begins 1.1 Numbering Read what your colleagues have posted. Encouraging lurkers (quiet ones) to contribute … Brief: … Purpose … Title … Next …"

In this edition each activity plan is set out as its own box, field by field.

B.2 Every formula had lost its two-dimensional layout

This is the worst case of the six documents processed so far. In the stored text layer each symbol is a separate object, and subscripts and superscripts sit on their own baselines, so a formula such as

$$C_t = \beta_1 + \beta_2 Y_t + u_t$$

was stored as the fragments `YC`, `ttt`, `21`, `IC`, `t`, `2`, `1`, `IIE`, … with no reliable order. For example, one line of the file reads

```
21YCttt ICY 22 ))(()( 2 tttt YEYCEC 0E E 0 0 t
```

where the page carries the model $C_t = \beta_1 + \beta_2 Y_t + u_t$, the identity $Y_t = C_t + I_t$, the deviation-form definitions $c_t = C_t - E(C_t)$ and $y_t = Y_t - E(Y_t)$, and the error assumptions $E(u_t) = 0$, $E\left(u_t^2\right) = \sigma^2$, $E\left(u_t u_{t'}\right) = 0$.

Repair. Every formula in this edition has been re-set in LaTeX and, wherever a result is quoted, the derivation re-worked so that it lands on the result the module states. The prose — which is intact — fixes the meaning of each symbol, so the reconstructions are determinate; where a page's algebra is genuinely ambiguous the reconstruction is flagged in a note at that point, and Appendix B.4 lists them.

B.3 Tables and figures

- The **coefficient tables** of Lessons 3 and 4 and the **parameter matrices** of Examples 4.5 and 4.6 survived only as loose numbers (e.g. "`1 0 0 1 0 0 2 1 0 0 0 3 0 0 1 0 0`"). They have been rebuilt as tables and matrices, and the determinants recomputed so that they give the module's own conclusions.
- The page images in the download (`bg1.png` … `bg33.png`) are **decorative only**: the university crest, the blue table-background of the title slides, and plain ruled boxes. No mathematical content exists in them, so nothing was lost by setting the mathematics from the text layer.
- The **cover slide** of Lesson 1 in the images carries the single word "SIMULTANEOUS EQUATIONS" over a blue banner; that is a slide title, not content.

B.4 Figures kept as printed, with the recomputed value in a note

Each of these is reproduced as the module has it, with the discrepancy explained in a shaded note at that point.

#	Module page	As printed	Recomputed
1	12	"the sample covariance of Y and u … as the sample size increases … implies that the estimate approaches the true population values"	the estimator is *inconsistent*: its probability limit is $\beta_2 + \mathrm{Cov}(Y,u)/\mathrm{Var}(Y) \neq \beta_2$
2	30–31	Example 4.2: $K = 6, M = 4, K - M = G - 1 = 2$, equation exactly identified	with the system as printed, $K = 5, M = 2$, $K - M = 3 > G - 1 = 2$, over-identified; and the variables listed on page 31 belong to no equation of that system
3	33	Example 4.4, equation (2): $M = 2, K - M = 4 > 3$ (over-identified)	the investment function contains I_t, Y_t, Y_{t-1}, so $M = 3$, $K - M = 3 = G - 1$ (exactly identified)
4	33	Example 4.4, equation (3): $M = 1, K - M = 5$	the tax function contains T_t, Y_t, so $M = 2, K - M = 4$; still over-identified
5	34	the tax function is numbered (3) but the table's fourth row is the identity	retained; the identity is not an econometric equation
6	36–38	the excluded-variable matrices are printed without their columns (e.g. "`0 1 0`", "`1 0 0 0`")	the sub-matrices of the coefficient table of Example 4.4 are given, and their determinants (0; −1; −1) reproduce the module's conclusions
7	39	reduced-form rank condition stated as "at least one **zero** determinant of order $G - 1$"	the rule as used in Example 4.6 requires a **non-zero** determinant; both the statement and the correct wording are given
8	21, 22	the "table of coefficients" of Lesson 3 is printed as a single run of digits	rebuilt as a table of β's and γ's
9	23–24	the direct/indirect/total effect decomposition is printed with the symbols detached from their labels	given as $\pi_{31} = \pi_{33} + \pi_{32}c_1$, i.e. total = direct + indirect
10	15	Lesson 2 opens with the heading "1.1 Introduction"	numbered 2.1
11	19	Lesson 3 opens with "1.1 Introduction"; §3.3 is followed by "3.4"	renumbered 3.1, and the subsection order made consecutive
12	27	Lesson 4 opens with "1.1 Introduction"	numbered 4.1
13	3	contents page lists "3.5 E-tivity" with no page number, and "3.6" twice	corrected in the lesson map of this edition
14	17	§5.2.1 is used both for "Identification and choice of estimation method" and inside the learning outcomes	the learning outcomes keep 5.2.1 and 5.2.2 as the module has them; the method sections are numbered 5.3, 5.3.1, 5.3.2 in this edition
15	27, 30, 31	"Defination", "Soultion.", "ansd", "differ betweenasymptotic"	"Definition", "Solution.", "and", "differentiate between asymptotic" — obvious slips normalised; the module's consistent "exogeneous/endogeneous" spelling is retained as printed
16	12, 18, 25, 42	the four E-tivities carry the module's field labels with their values in the wrong cells (e.g. a *Purpose* value printed under *Brief summary of overall task*)	fields assigned by meaning
17	52	Lesson 5's second example stops its solution at the instruction "Divide (10) with (7)", with the symbols detached from their slots	the three operations are carried out, and $\widehat{\alpha}_1, \widehat{\alpha}_2$ completed from equations (7) and (8)
18	55	the structural model of §6.2.1 printed with its subscripts as loose numerals ("13131122111", "24233221212")	re-set with consecutive subscripts, keeping the regressors the page's reduced forms imply
19	56	Assumption 3 of 2SLS justified with the sentence "in the ILS method applies the OLS to the reduced form equations", copied from Lesson 5	corrected to the first stage of 2SLS
20	55, 59	the second-stage normal equations of §6.2.3 survive only as detached symbols	the three normal equations assembled from them

#	Module page	As printed	Recomputed
21	68–70	Examples 7.1 and 7.2 print their systems as symbol runs (“zy”, “zzyy”, “β_1”, “γ_1”…)	systems restored to the structure their solutions presuppose; three normal equations rebuilt
22	2–5, 55, 66	contents pages list Lesson 6 as IV and Lesson 7 as 2SLS; the body teaches 2SLS first	body order followed and stated in the lesson map
23	53, 63, 70	the e-tivity of Lesson 6 is numbered “7.1”, and those of Lesson 7 are numbered “6.4”/“6.5”/“6.6”	renumbered to the lesson each belongs to; the module’s own numerals kept inside the field values
24	66–67	Assumption 3 of the IV method carries four sub-items and then “v) ASSUMPTION 4”, “vi) ASSUMPTION 5”	the five assumptions numbered 1 to 5
25	53, 64, 71	the exercise systems of Lessons 5, 6 and 7 printed as digit runs with the variable key detached	systems restored from the symbols and the variable key on the page; noted at each point

B.5 What was *not* altered

The lesson prose is the module’s own wording. The module prints the same five-item reference list at the end of every lesson. Where a lesson’s list is identical to the one already given (Lessons 2, 3 and 4), this edition points back to §2.6, which carries the full list with ISBNs, rather than printing the same page four times; Lesson 1 carries the list in full, and Lessons 5, 6 and 7 — which are the ones being consulted when a student is working on estimation — carry it in full as the module prints it.

APPENDIX C — SOURCE PAGE MAP

Page	Content
1	Title page: Kenyatta University, Digital School of Virtual and Open Learning, SST 404 Econometrics II, KU Press 2020
2–5	Table of contents (lessons 1–7 and their sections)
6	Course description: purpose, expected learning outcomes, course content, mode of delivery, assessment, core reading
7	Recommended reading materials
8	Lesson 1: introduction, learning outcomes, §1.3 simultaneous equation relationships begins
9	Demand equation and assumptions; income determination model; consequences of simultaneous equations
10	Covariance of Y and u; commodity demand model
11	Assumptions of the commodity model; price equation; covariance of P and u
12	Conclusion of Lesson 1 example; Activity 1.4 begins
13	§1.6 exercises; Activity 1.4 fields
14	§1.7 references
15	Lesson 2: introduction, outcomes, §2.3 equation bias, deviation form, normal equations
16	Small-sample bias; probability limit of $\widehat{\beta}_2$
17	Asymptotic bias; equation (2.4)
18	Activity 2.4; §2.5 exercises
19	Exercise continued; §2.6 references; Lesson 3: introduction
20	§3.3 structural equations form; the model and its β/γ notation
21	Arrangement of the system for identification; table of coefficients; §3.4 reduced form
22	Reduced form derivation, Steps 1–3
23	Comparison of coefficients; direct and indirect effects
24	Indirect effect traced; Activity 3.6
25	§3.6 exercises; Activity 3.6 fields
26	Exercise continued; §3.7 references
27	Lesson 4: introduction, outcomes, §4.3 identification
28	Exactly/over/under-identified; Example 4.1 market model
29	Unidentified model; §4.4 rules; order condition
30	Order condition in symbols; Example 4.2
31	Example 4.2 figures; Example 4.3
32	Examples 4.3 and 4.4; coefficient table
33	Identifiability of equations (1)–(3); rank condition rule
34	Rank condition; Example 4.5; matrix Q_1
35	Example 4.5 continued; tables of excluded variables
36–38	Rank condition for equations (2) and (3); determinants and conclusions
39	§4.4.2 reduced model approach; Example 4.6 begins
40	Reduced form of Example 4.6; order condition for equation (1)
41	Rank condition; Tables 1 and 2; exercise on equations (2) and (3)
42	Activity 4.4 fields
43	§4.5 exercises (two systems)

Page	Content
44	§4.6 references
45	Lesson 5: introduction; the two groups of estimation methods
46	Choice of method; ILS; assumptions 1 and 2
47	Assumptions 2(i)–(vi); Assumption 3; the ILS process, Steps 1–4
48	Step 5; §5.5 Example 1 — identification of the Keynesian model
49	Reduced form and the two forms of the model; Steps 2–5
50	Parameter equations solved; §5.2.2 properties of ILS estimates; Example 2 begins
51	Example 2: identification, reduced form (3) and (4)
52	Example 2 continued: parameter equations (5) and (6), estimates (7)–(12), solution by division and substitution
53	§5.6 E-tivity 5; §5.7 exercises (system in Y_1, Y_2, I, M)
54	Variable key for the exercise; §5.8 references
55	Lesson 6: introduction, learning outcomes, §6.2.1 assumptions of 2SLS; the model (6.1)–(6.2); reduced forms and estimated forms (6.3)–(6.4)
56	Assumption 2 (i)–(vi); Assumption 3 begins
57	Assumption 3 (i)–(vi) for the reduced form
58	Assumptions 4–6; outline of the 2SLS procedure, Steps 1–2
59	Steps 3–5; §6.2.2 application — the two-equation system and its coefficient table
60	Order and rank conditions for equation 1; order condition for equation 2
61	Rank condition for equation 2; choice of method; estimation of equation 1 by 2SLS: Steps 1–2 and the normal equations
62	Exercise on equation 2; §6.2.3 properties of 2SLS estimators; advantages
63	§6.3 E-tivity 6
64	§6.4 exercise (two systems, including the money market); §6.5 references begin
65	References concluded
66	Lesson 7: §7.1 the IV method; §7.2 learning outcomes; §7.2.1 assumptions 1–3
67	Assumptions 4–5; the method and its outline, Steps 1–5
68	Example 7.1: choice of instrument, Steps 1–3, the IV estimator
69	Substitution route; Example 7.2: equation (7.3) analysis and the three instruments
70	Normal equations (7.5)–(7.7); exercise on equation (7.4); §7.3 E-tivity 7
71	§7.4 exercise; §7.5 references; “END OF MODULE”
72	References concluded